CHAPTER 53
FUSELAGE

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ILLUSTRATIONS

CHAPTER 53
FUSELAGE

53-00-00 INTRODUCTION

The Dash 8 - Series Q400 fuselage is a semi-monocoque structure with skin that is reinforced by frames and longitudinal stringers.

The fuselage consists of the five main sections as follows:

- Nose
- Forward center
- Middle center
- Aft center
- Aft tail.

The basic structure is a conventional all metal construction utilizing high strength aluminum alloys (2024 and 7075), frames, stringers and chemical milled skin.

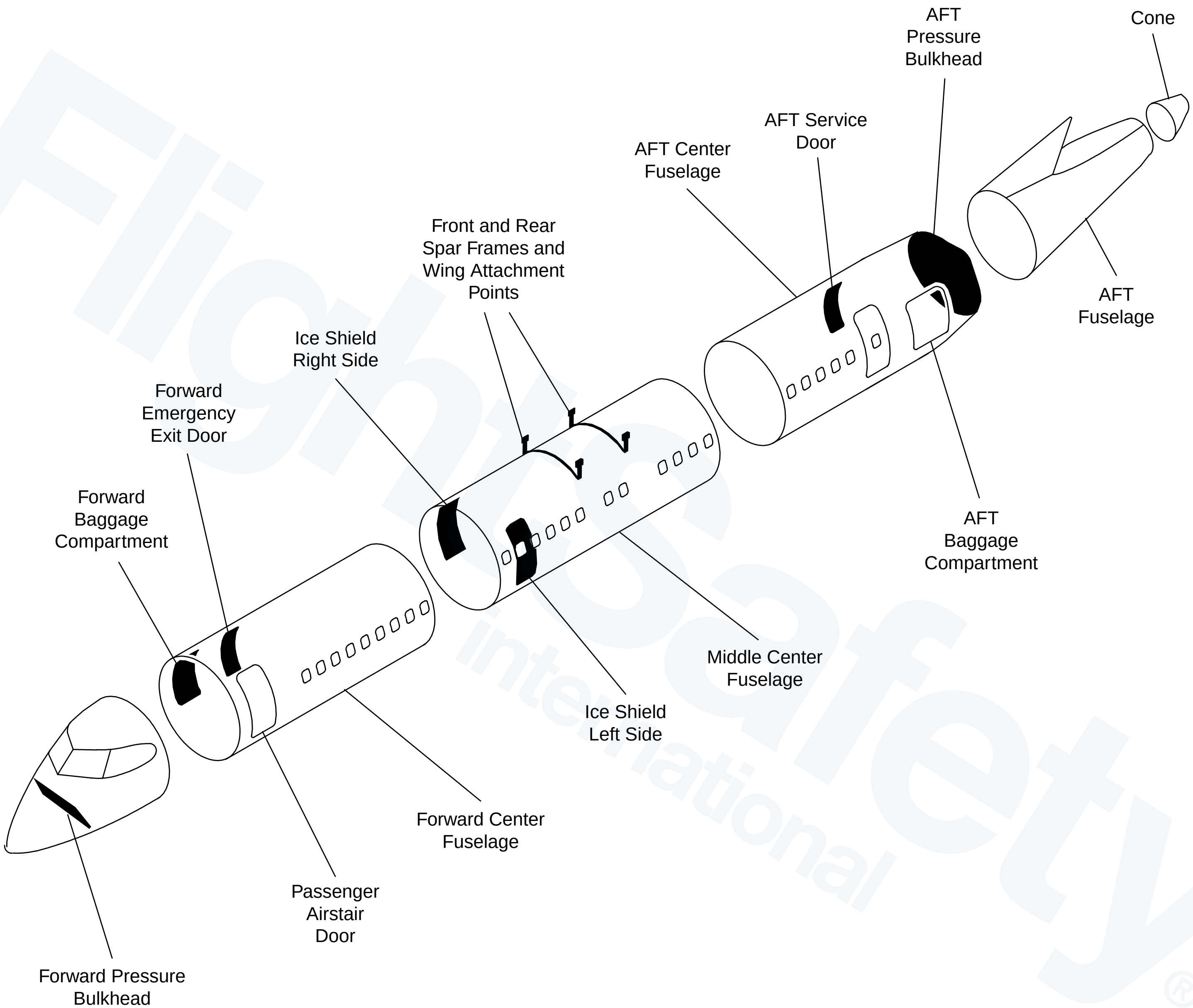

Figure 53-1. Fuselage Structural Identification

GENERAL

Refer to Figure 53-1. Fuselage Structural Identification.

NOSE SECTION

The nose section extends rearward to the fuselage forward center section, from station X -155.000 to X -19.850.

It contains the flight compartment, that is separated from the main cabin by a bulkhead with a lockable door.

A removable escape hatch is provided in the canopy roof of the flight compartment.

The windshield panels are made of laminated glass construction, while the side panels are of laminated plastic construction. Both windshield and side panels are stressed to take pressure from inside and withstand birdstrikes.

The forward pressure bulkhead is located at station X -111.000. The area forward of the bulkhead encloses the unpressurized equipment compartment and supports the nose cone and radar unit.

The nose landing gear and nosewheel compartment are located below the equipment compartment.

FORWARD CENTER FUSELAGE

The forward center section extends from the nose section, station X -19.775, to the mid-center section, station X 234.475.

It is a constant circular cross section of 106 inches (2692 mm) outer diameter, with a flattened bottom of larger radius.

Extensive use is made of bonded frames, stringers and window reinforcements to the skin, to achieve minimum weight, permit a flush bounded exterior and to stop crack propagation.

The forward passenger door is on the left side of the forward center fuselage.

The forward baggage compartment is at the front right side of this section and is accessible through an external door on the right side of the fuselage or internal door from cabin.

A type II/III emergency exit door is installed on the forward right side of the fuselage, just aft of the forward baggage compartment door. The emergency exit door is constructed of an upper and lower door. In the event of an emergency ditching procedure, the lower door is kept closed during passenger evacuation.

MIDDLE CENTER FUSELAGE

The middle center section extends from the forward center section, station X 234.475, to the aft center section, station X 566.025.

Its construction is similar to the forward center fuselage.

The wing structure is attached to fittings on the middle center section with tension bolts.

The floor loads are supported by the seat rails and frames. The floor structure can stabilize the frames in the event of a wheel up landing.

Ice shields are installed on the left and right side of the middle center fuselage.

These ice shields are installed on the plane of the propellers, to protect the fuselage pressure shell from damage due to the impact of ice thrown from the propellers.

AFT CENTER FUSELAGE

The aft center section extends from the middle center section, station X 566.025, to the aft/tail section, station X 829.548/836.452.

Its construction is similar to the forward center fuselage.

The aft passenger entry door is located on the aft side of the aft center fuselage.

The service door is located opposite of the passenger entry door.

The aft baggage compartment is at the rear of this section and is accessible through an external door on the left side of the fuselage.

AFT/TAIL FUSELAGE

The aft/tail fuselage section extends from the aft center section, station X 829.548/836.452, to the rear of the aft/tail section, station X 1060.670 including tail cone.

The vertical stabilizer and dorsal fin are a part of the aft/tail fuselage. The lower part of the three vertical stabilizer spars extend down to make the mainframe of the aft/tail fuselage.

The area between the front and center frame is an equipment bay for the air cycle machine and other equipment.

Access to the equipment bay is through an access door located on the underside of the aft/tail fuselage.

TRAINING INFORMATION POINTS

The fuselage has allowable damage points that follow:

- Oversized rivet substitution to remove damage on the aircraft fuselage skins around full head countersunk solid rivets (B0205017 ref) due to lightning strike or other types of damage
- Temporary continued flight operation with multiple lightning strike damage locations to fuselage metal skin bay panels
- Supplementary (supplement to SRM limits) skin damage blend-out limits for any damage such as minor scratches, nicks, gouges, lighting burns, or corrosion
- Generic permanent repair for minor dents and scratches on the pressurized fuselage and door metallic skins between the Forward Pressure Bulkhead (FPB) and the Rear Pressure Bulkhead (RPB)
- Generic temporary repair for damaged anchor nuts at any fuselage-mounted antenna.

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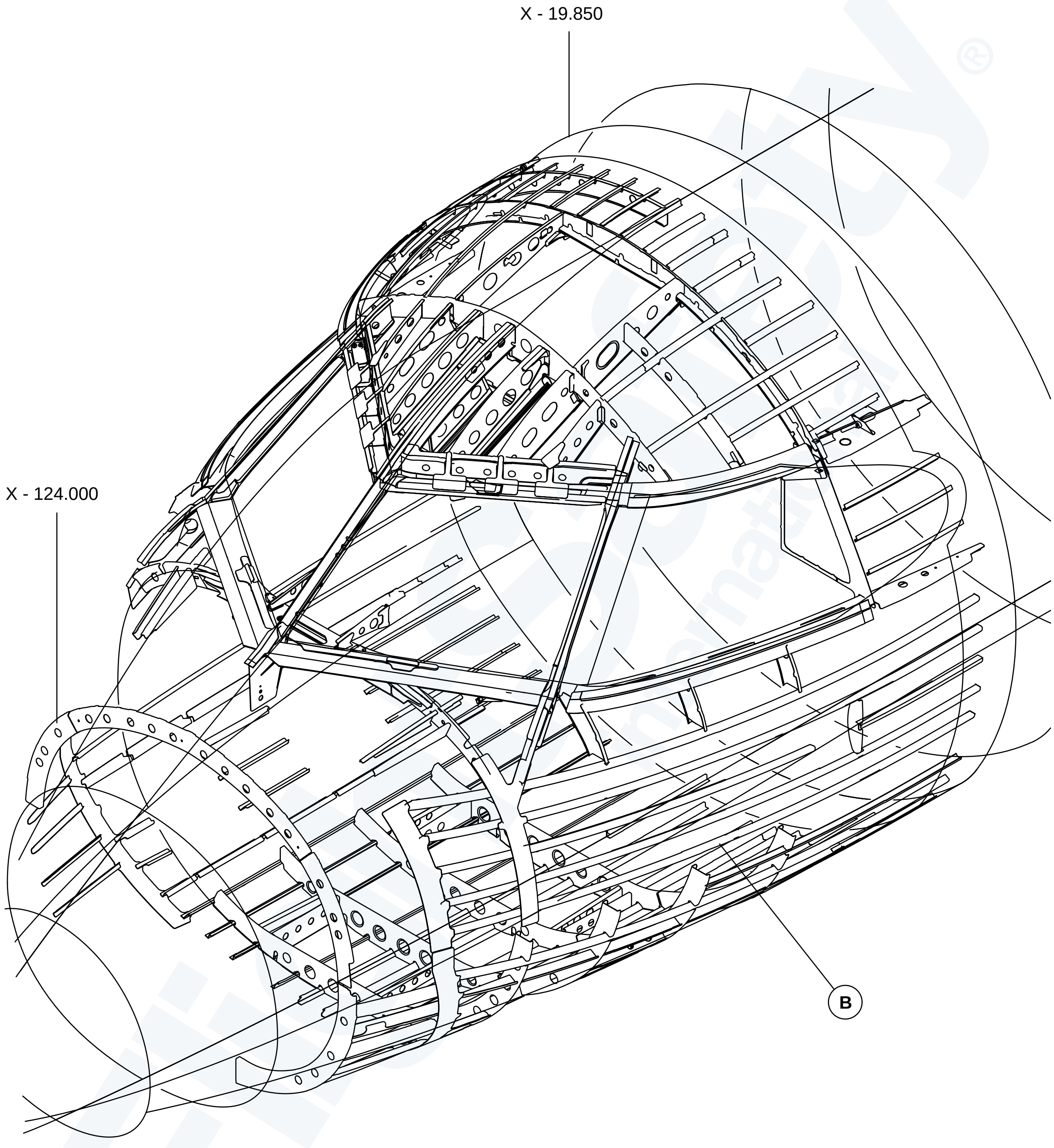

STRINGERS, LHS FRONT FUSELAGE

Figure 53-2. Nose Fuselage - LHS

53-10-00 NOSE FUSELAGE (STATIONX-178.000 TO X-18.250)

INTRODUCTION

The nose section extends rearward to the fuselage forward center section, from station X -155.000 to X - 19.850 along the longitudinal X-axis.

GENERAL

Refer to Figure 53-2. Nose Fuselage - LHS.

The nose section contains the flight compartment which is separated from the main cabin by the pilot's bulkhead at the fuselage station X -39.000. The pilot's bulkhead contains a lockable door - between the flight compartment and the passengers' (main) cabin - and the front jacking point at the bottom of the bulkhead.

The flight compartment contains the forward pressure bulkhead. The pressure bulkhead consists of two parts:

- The upper pressure bulkhead, which is the vertical portion of the pressure bulkhead at the fuselage station X -111.000.
- The lower pressure bulkhead, which is inclined rearward between fuselage stations X -111.000 at Z 112.000 (the flight compartment floor level) to X -102.000 at the bottom skin.
- The area forward of the bulkhead encloses the unpressurized equipment compartment and supports the nose cone and radar unit.

The nose landing gear and nosewheel compartment are located below the equipment compartment.

The flight compartment escape hatch is a removable hatch in the canopy roof and also provides a means of ventilation when the aircraft is on the ground.

The windshield panels are made of laminated glass construction, while the side windows panels are made of laminated plastic construction. Both windshield and side windows are designed to withstand cabin pressurization and the windshield is designed to withstand bird strikes impact.

The nose fuselage structure has the following components:

- Stringers
- Frames
- Intercostals
- Forward pressure bulkhead
- Pilots (flight compartment) bulkhead
- Fuselage skin
- Joint installation, front to forward center fuselage
- Windshield and side window frame structure
- Floor support structure, flight compartment
- Flight compartment floor panels
- Flight compartment seat rail
- Flight compartment secondary structures
- Nose landing gear support structure.

53-10-07 WINDSHIELD AND SIDE-WINDOW SURROUND STRUCTURE

Refer to Figure 53-3. Flight Compartment Windshield and Side Window Surround Structure - Detailed Visual Inspection.

TIP: DVI Inspection of opportunity (When you have the windshield out).

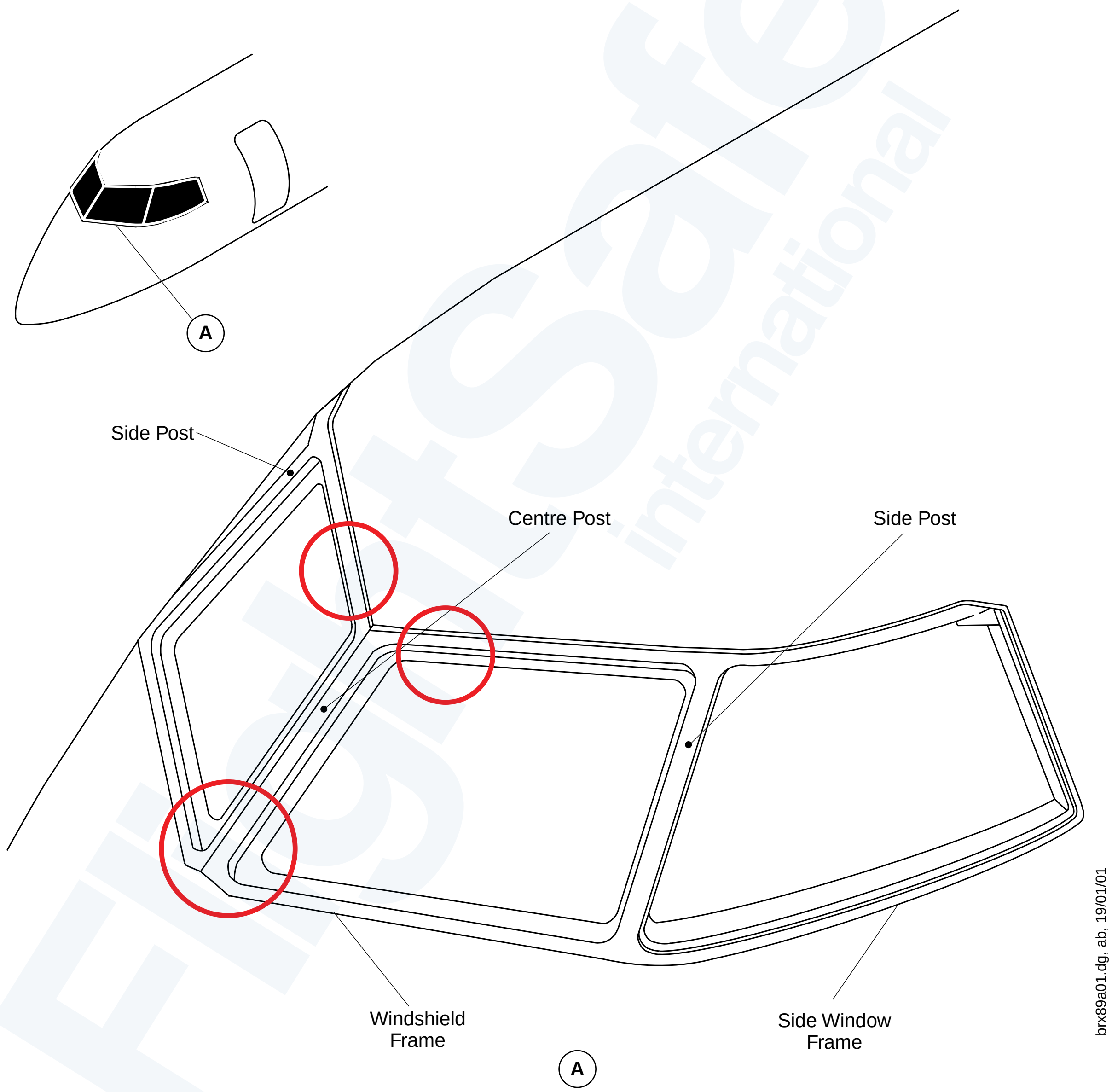

Figure 53-3. Flight Compartment Windshield and Side Window Surround Structure - Detailed Visual Inspection

53-10-01 SIDE WINDOW ATTACHMENT SCREWS

Refer to Figure 53-4. Flight Compartment Side Window.

TIP: GVI Inspection and Torque Check (1200 FH or 12 Mo.)

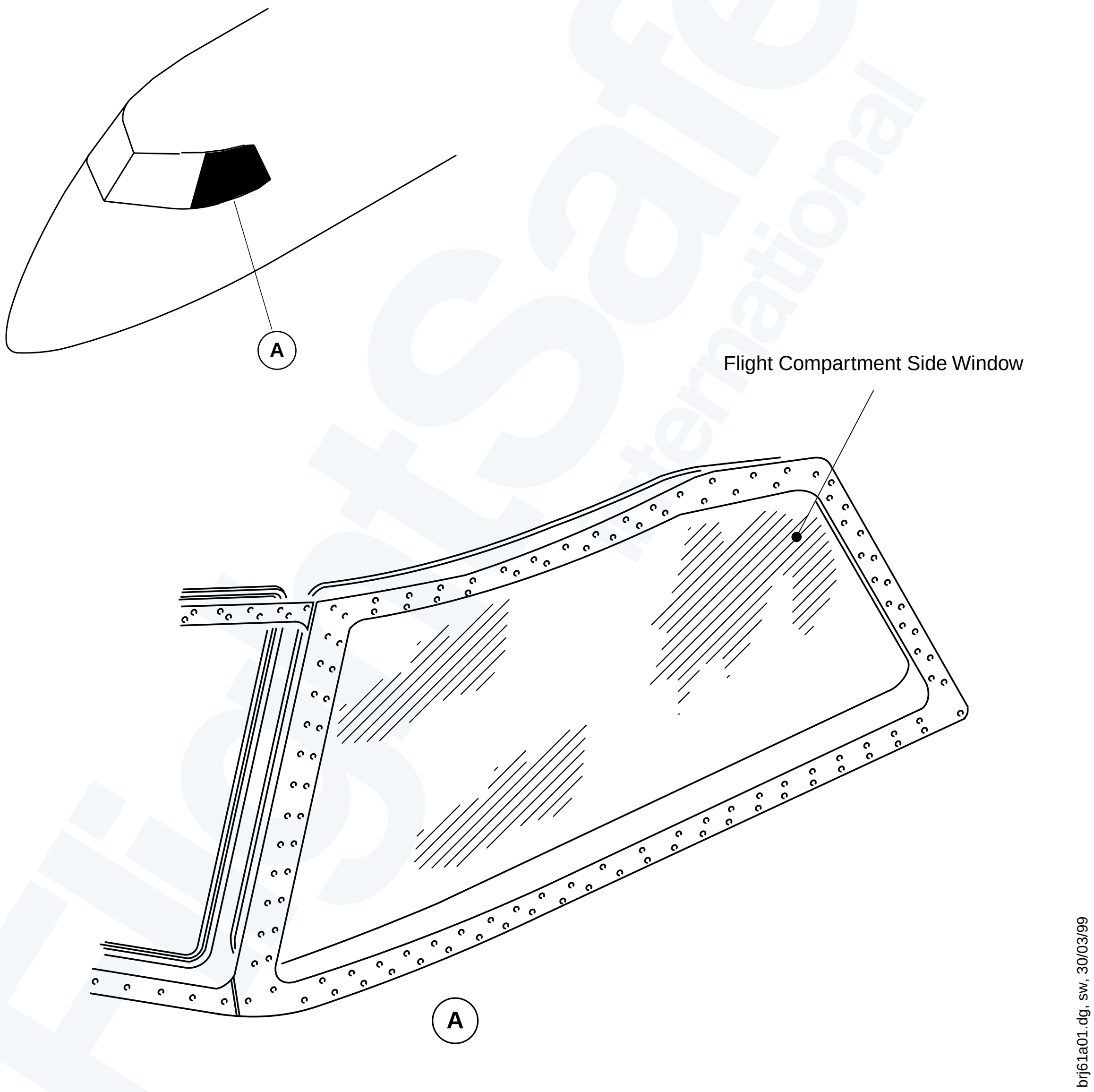

Figure 53-4. Flight Compartment Side Window

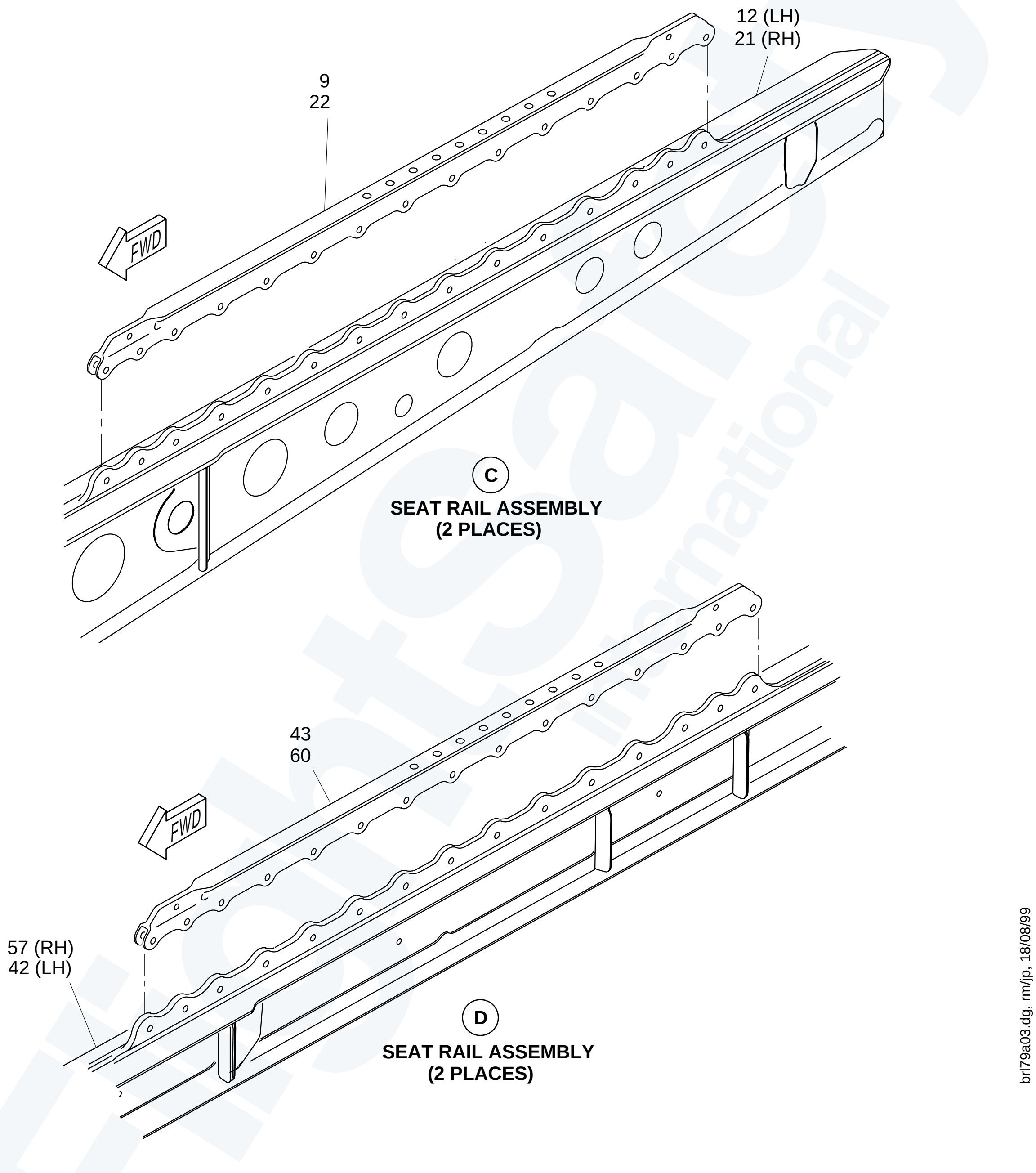

Figure 53-5. Flight Compartment Seat Rail

Flight Compartment Seat Rail

NOTE

Refer to:

- Figure 53-5. Flight Compartment Seat Rail.
- Figure 53-6. Flight Deck Manual Stowage.

No repair is allowed to the pilot and co-pilot’s seat rail.

The only repair allowed is, wear and blendout limits for the pilot and co-pilot’s stainless steel caps.

Seat rails are replaceable without structural re-work (NAS Bolts, nuts and washers).

Figure 53-6. Flight Deck Manual Stowage

53-10-67 NOSE LANDING GEAR SUPPORT STRUCTURE

Refer to Figure 53-7. Nose Landing Gear Support Structure.

TIP: GVI inspection (every 6 years)

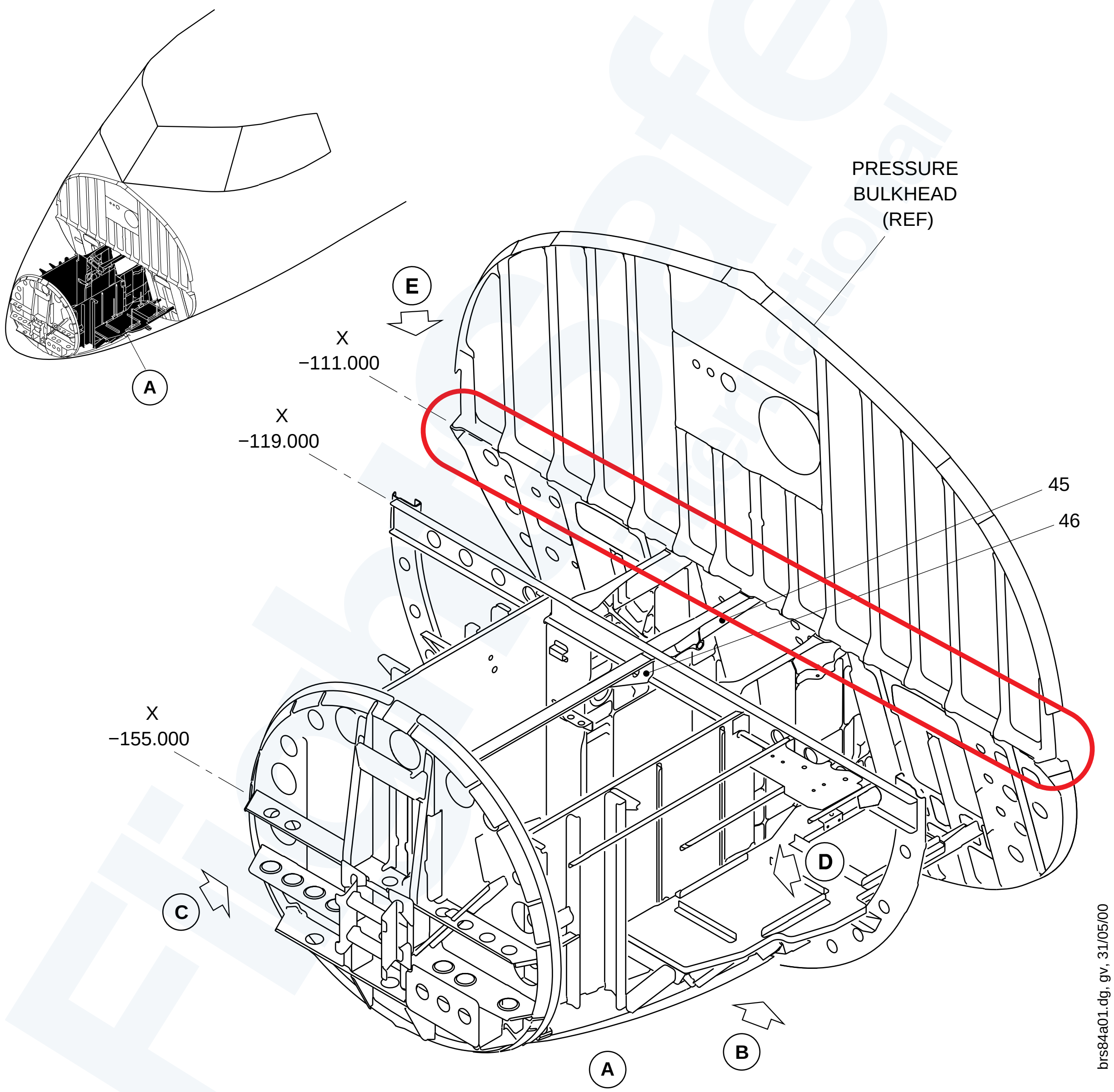

Figure 53-7. Nose Landing Gear Support Structure

Radome

Refer to Figure 53-8. Radome.

The radome is located on the nose of the aircraft.

The radome is made of fibreglass with a honeycomb core and lightning strip diverter strips. The bottom of the radome has two drain holes.

The radome gives protection for the weather radar and glideslope antennas. It also helps with the aerodynamic smoothness of the aircraft. The radome does not interfere with the operation of the glideslope and weather radar.

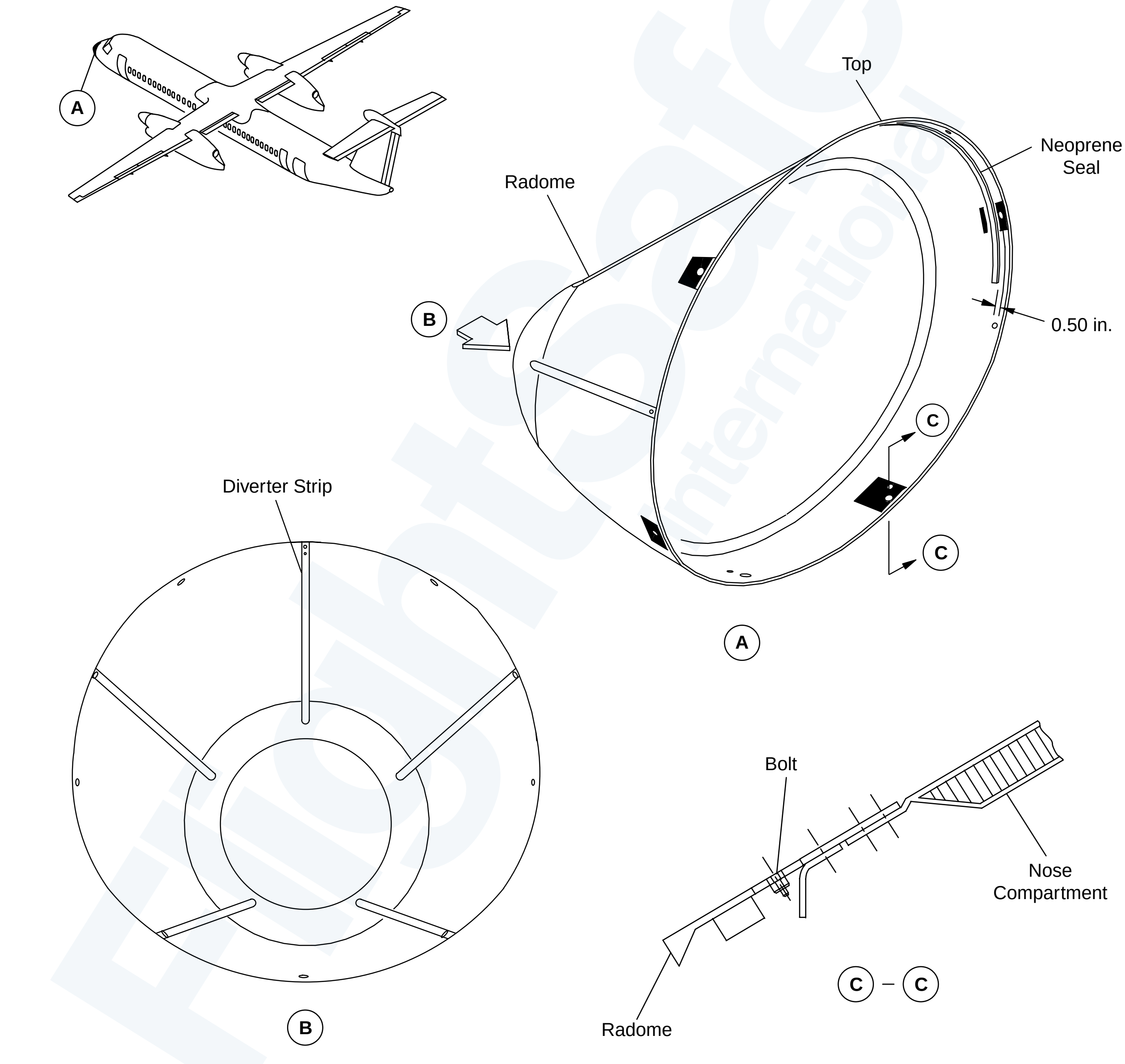

Figure 53-8. Radome

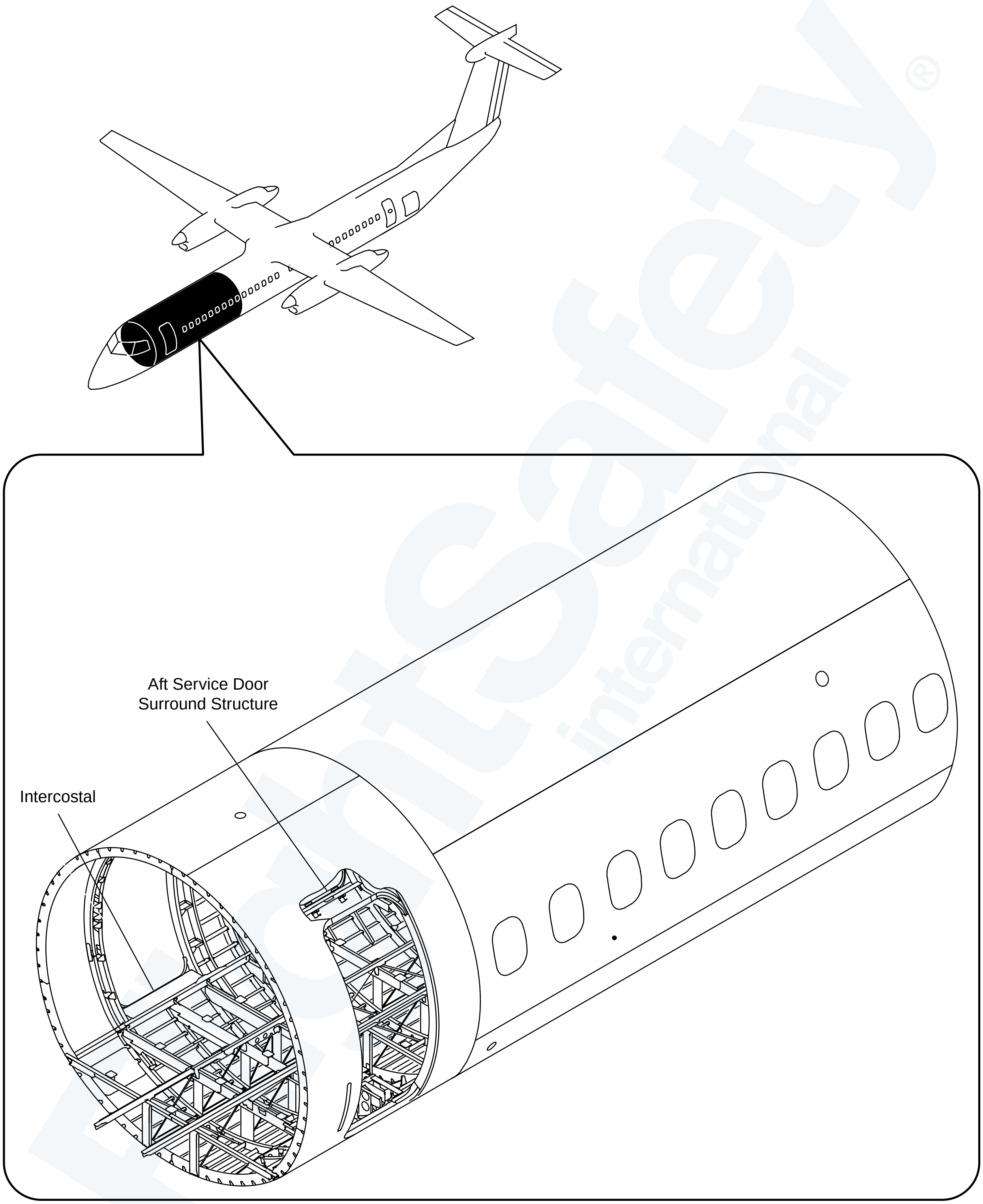

Figure 53-9. Forward Center Fuselage

53-20-00 FORWARD CENTER FUSELAGE (STATION X-18.250 TO X 236.000)

INTRODUCTION

The forward center section extends rearward to the fuselage forward center section, from station X -19.775 to X 234.475 along the longitudinal X-axis.

GENERAL

Refer to Figure 53-9. Forward Center Fuselage.

The forward center structure has the components that follow:

- Stringers
- Frames
- Intercostals
- Emergency exit door surround structure
- Passenger door surround structure
- Forward baggage door surround structures
- Fuselage skin
- Joint installation, center fuselage, forward center to middle center fuselage
- Floor support structure
- Passenger seat rail and drain plugs.

NOTE

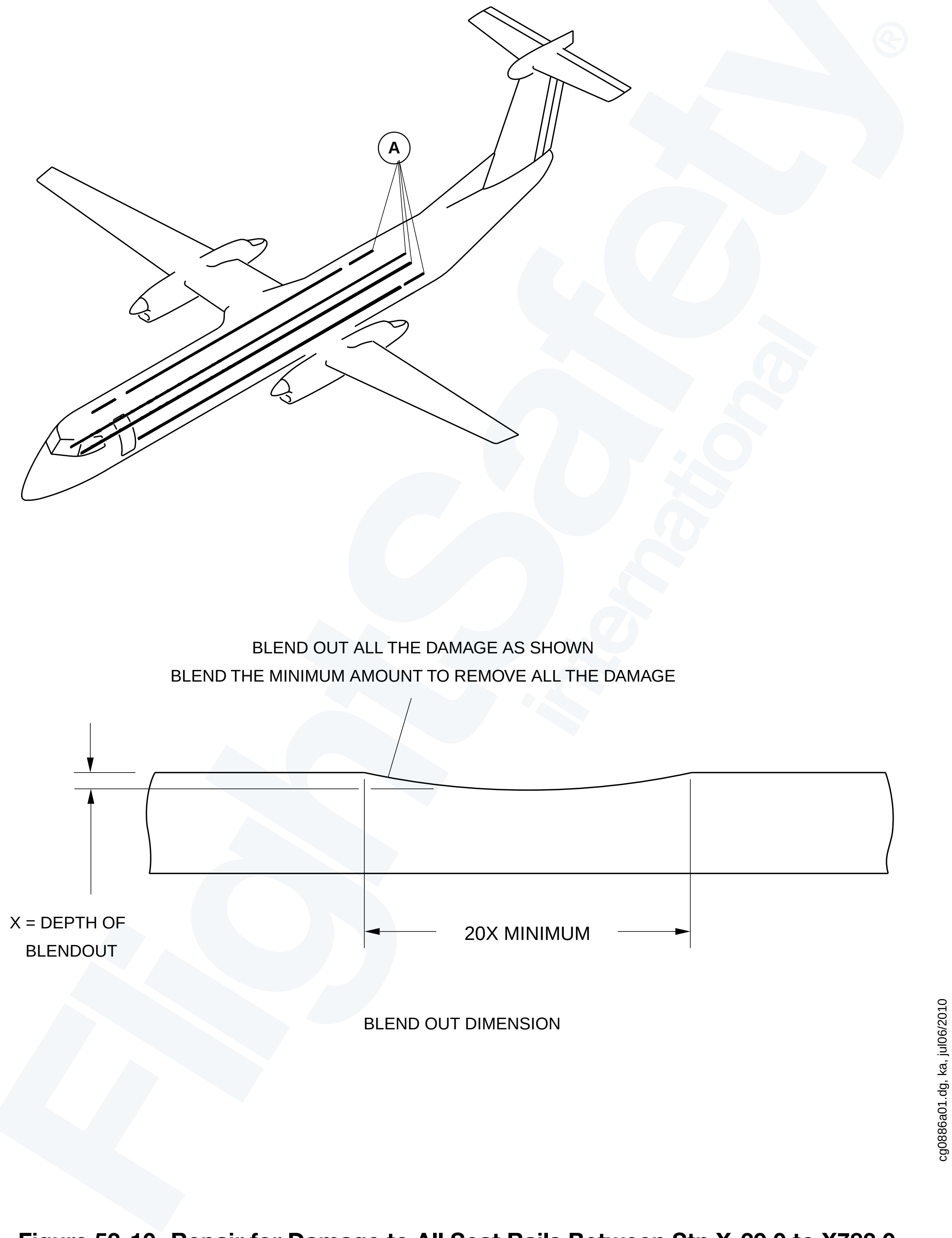

Figure 53-10. Repair for Damage to All Seat Rails Between Stn X-39.0 to X783.0

53-20-49 PASSENGER SEAT RAILS

Refer to:

- Figure 53-10. Repair for Damage to All Seat Rails Between Stn X-39.0 to X783.0.
- Figure 53-11. Underfloor Cabin.

The seat rails provide seat retention, while providing options for cabin configurations.

The seat rails can be replaced IAW the SRM but require the fuselage be jig supported and they are installed with structural fasteners.

There is also permanent repairs provided in the SRM if the damage is within limits.

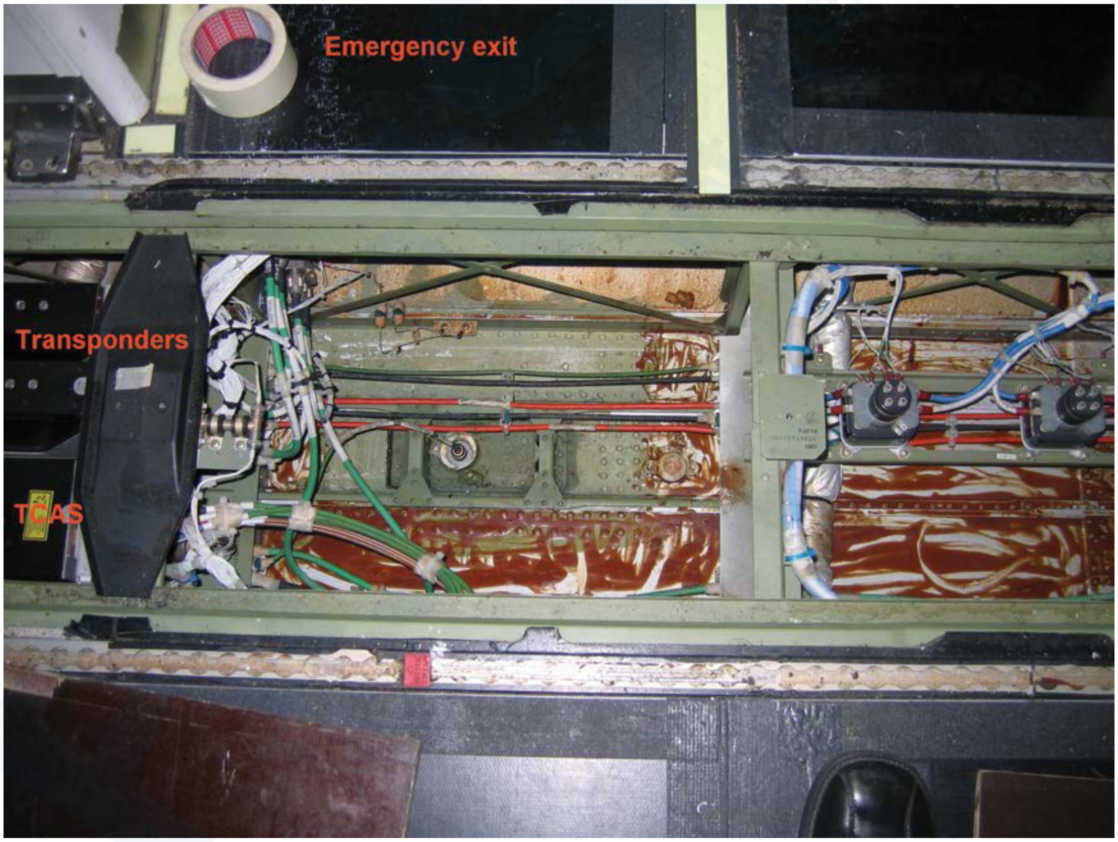

Figure 53-11. Underfloor Cabin

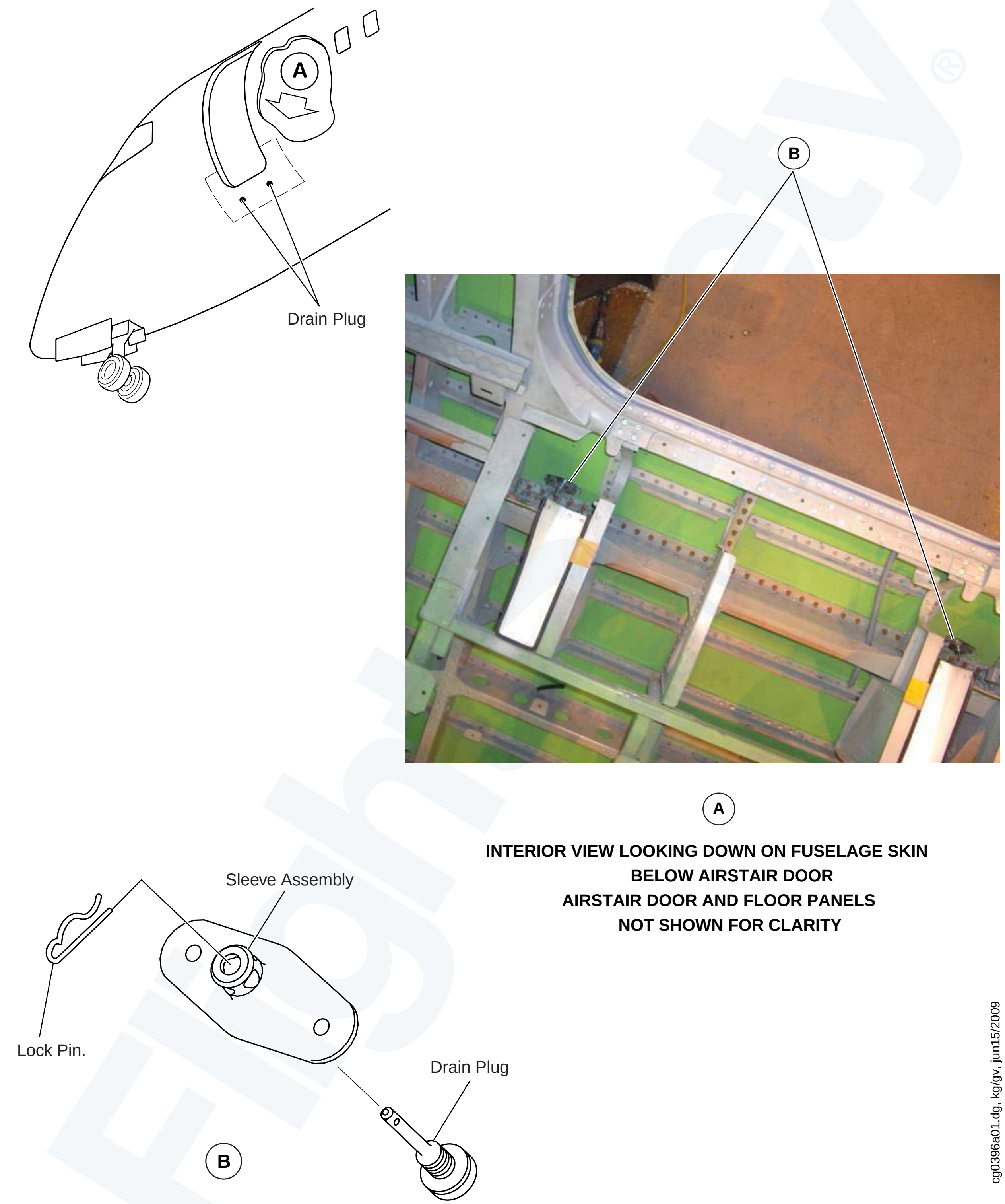

Figure 53-12. Airstair Door Threshold Drain Plug - Removal/Installation

53-20-23 AIRSTAIR DOOR THRESHOLD DRAIN PLUGS

Refer to:

- Figure 53-12. Airstair Door Threshold Drain Plug - Removal/Installation.
- Figure 53-13. Airstair Door Threshold Drain Plugs - Opening/Closing.

There are two drain plugs located below the airstair door threshold. The drain plugs are opened and closed to allow for draining of the water trapped in the fuselage airstair door threshold area.

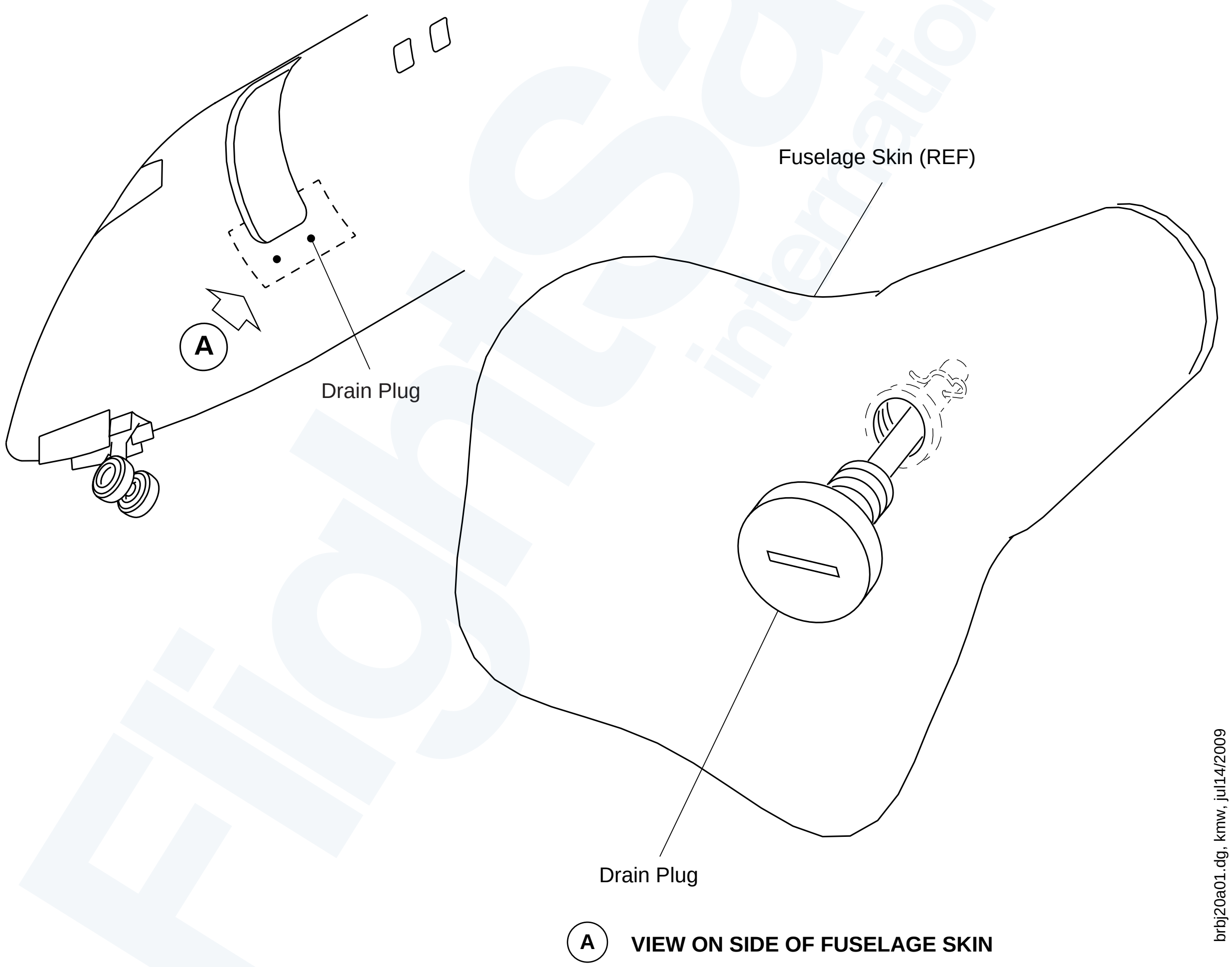

Figure 53-13. Airstair Door Threshold Drain Plugs - Opening/Closing

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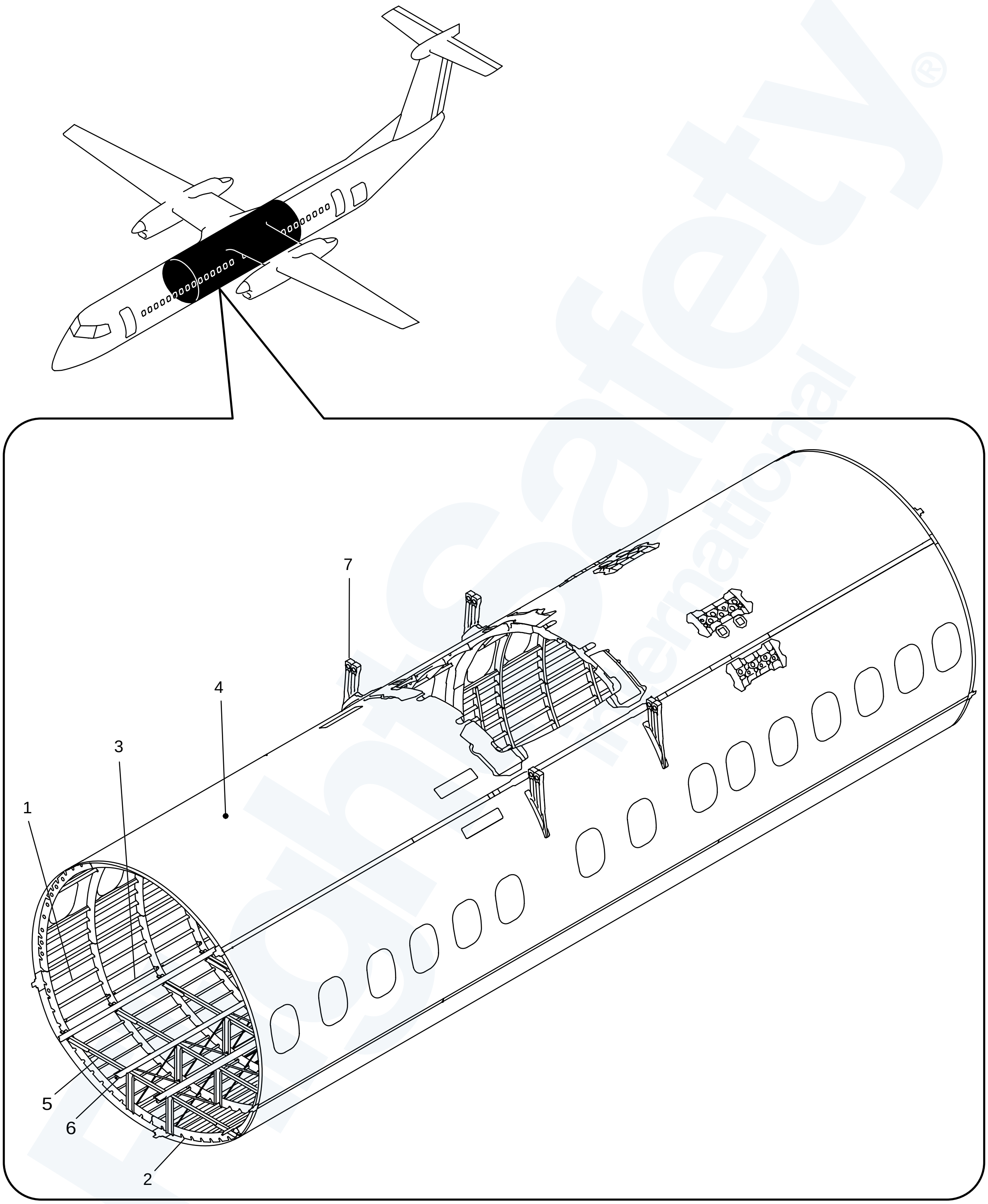

Figure 53-14. Middle Center Fuselage (X 234.475 to X 566.025)

53-30-00 MIDDLE CENTER FUSELAGE

INTRODUCTION

The middle center section extends rearward to the fuselage aft center section, from station X 234.475 to X 566.025 along the longitudinal X-axis.

GENERAL

Refer to Figure 53-14. Middle Center Fuselage (X 234.475 to X 566.025).

The middle center structure has the components that follow:

1. Stringers
2. Frames
3. Intercostals
4. Fuselage skin
5. Floor support structure
6. Passenger seat rails
7. Fuselage to wing support structure.

NOTE

COMPONENT DESCRIPTION

Middle Center Fuselage Access Panels

Refer to Figure 53-15. Middle Center Fuselage Access Panels.

The middle center fuselage section has the access panels as seen on the illustration.

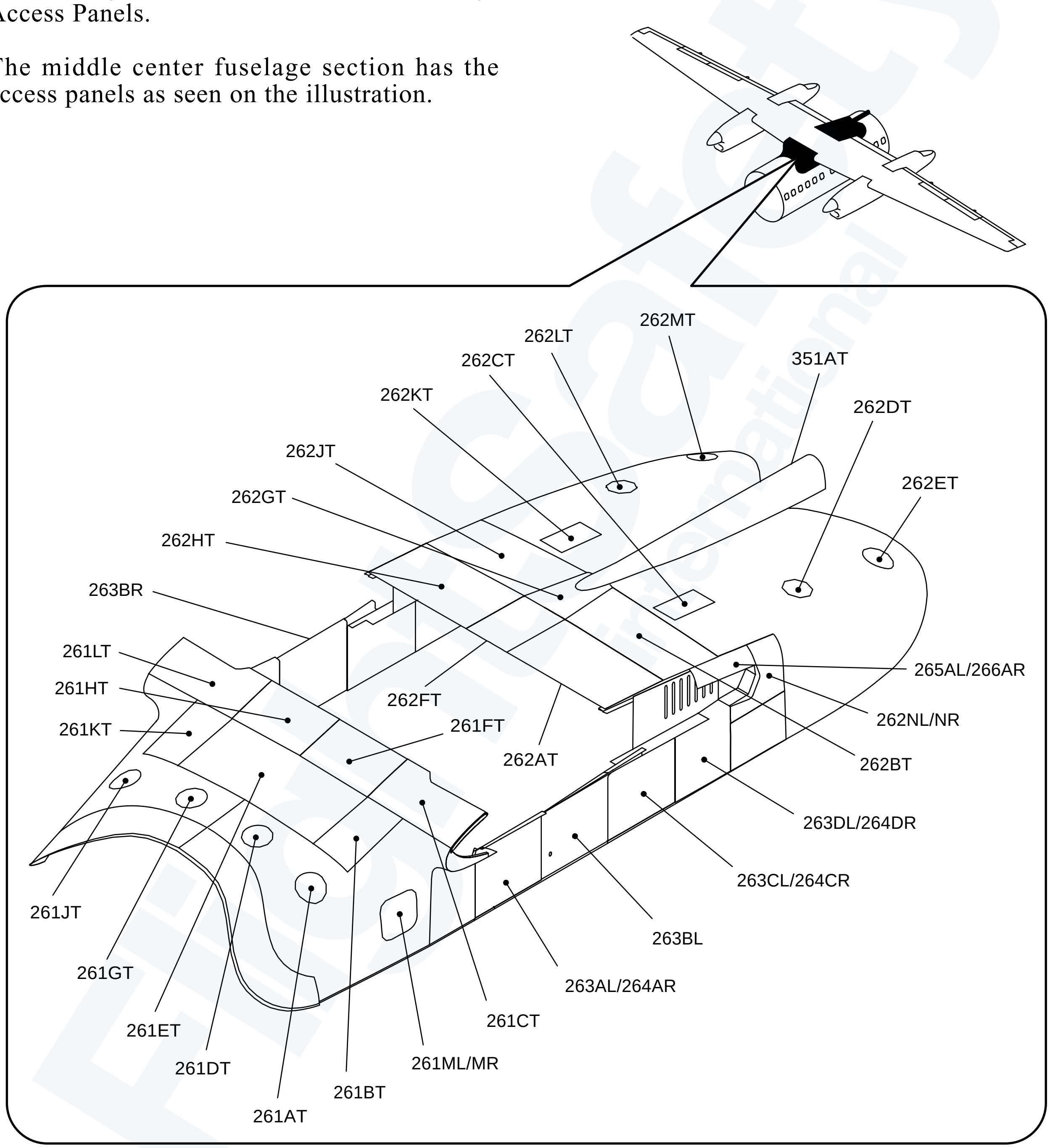

Figure 53-15. Middle Center Fuselage Access Panels

Ice Shields

Refer to Figure 53-16. Ice Shields.

The ice shields are installed on the right and left sides of the center fuselage.

An anti-abrasion coating is applied underneath the fuselage ice shield panels.

Aircraft without anti-abrasion coating are to be inspected under the left and right ice shields every 1000 hours or less for evidence of primer abrasion and skin corrosion.

Any corrosion or wear found is to be reported to Bombardier Aerospace. Any bare metal should be alodined and epoxy primed.

Aircraft may be flown for up to 5000 hours before removing the ice shields and brush applying two coats of F35 epoxy coating to give a dry film thickness.

The ice shields prevent damage to the fuselage caused by propeller anti-icing.

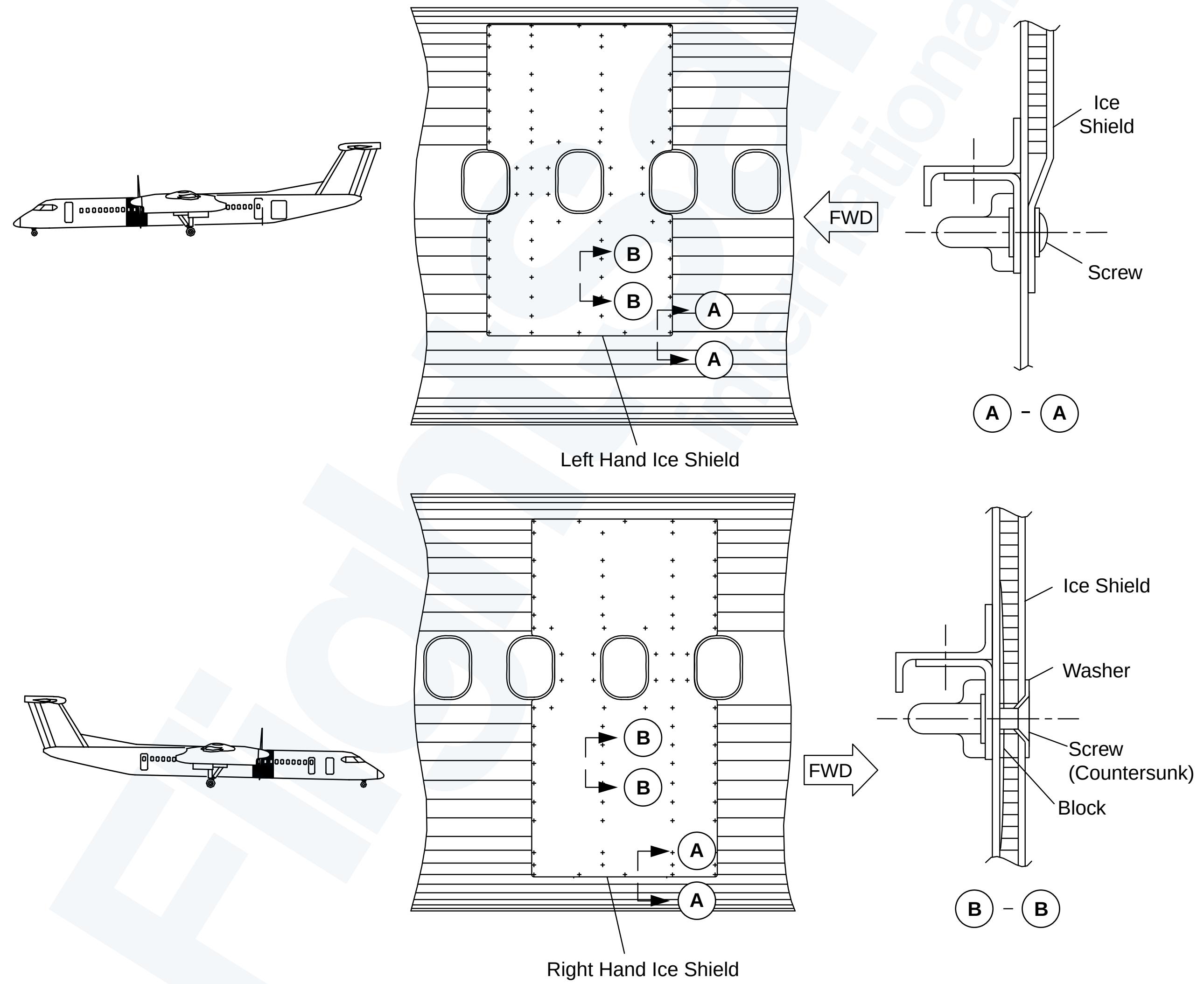

Figure 53-16. Ice Shields

Wing to Fuselage Attachment Bolts

Refer to Figure 53-17. Wing to Fuselage Attachment Bolts.

Tension bolts attach the wing to the fuselage. The front and rear spars of the center wing structure are attached to fittings on the fuselage with sixteen tension bolts.

The fittings are made of AMS5659 corrosion resistant steel.

Do not remove more than one tension bolt and only one location at a time.

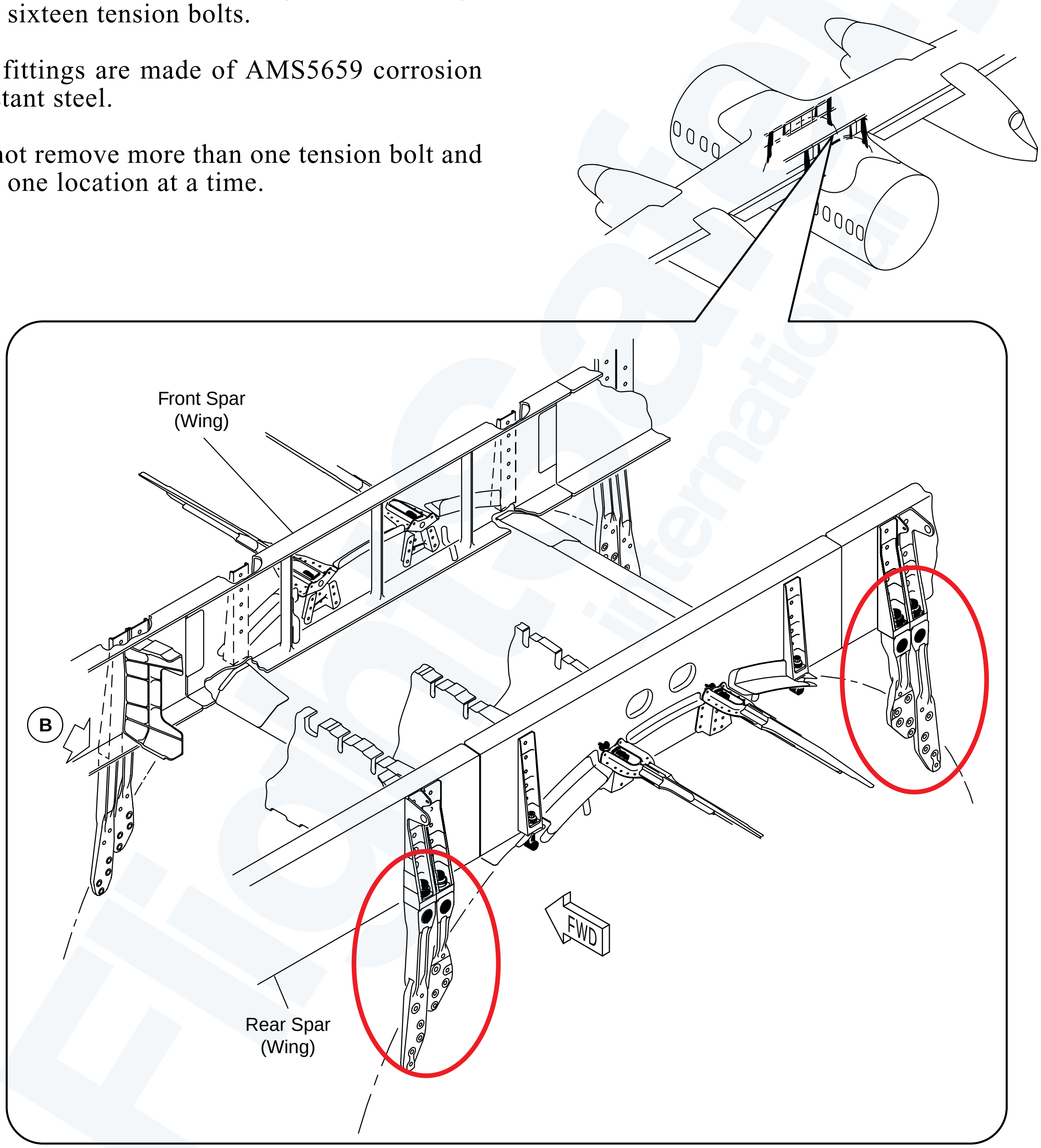

Figure 53-17. Wing to Fuselage Attachment Bolts

Wing to Fuselage Joint Struts

Refer to Figure 53-18. Wing to Fuselage Joint Struts.

Four bolts on the outer side of the fuselage and eight bolts on the inner side of the fuselage are installed at each spar.

The struts are made of AMS5659 corrosion resistant steel.

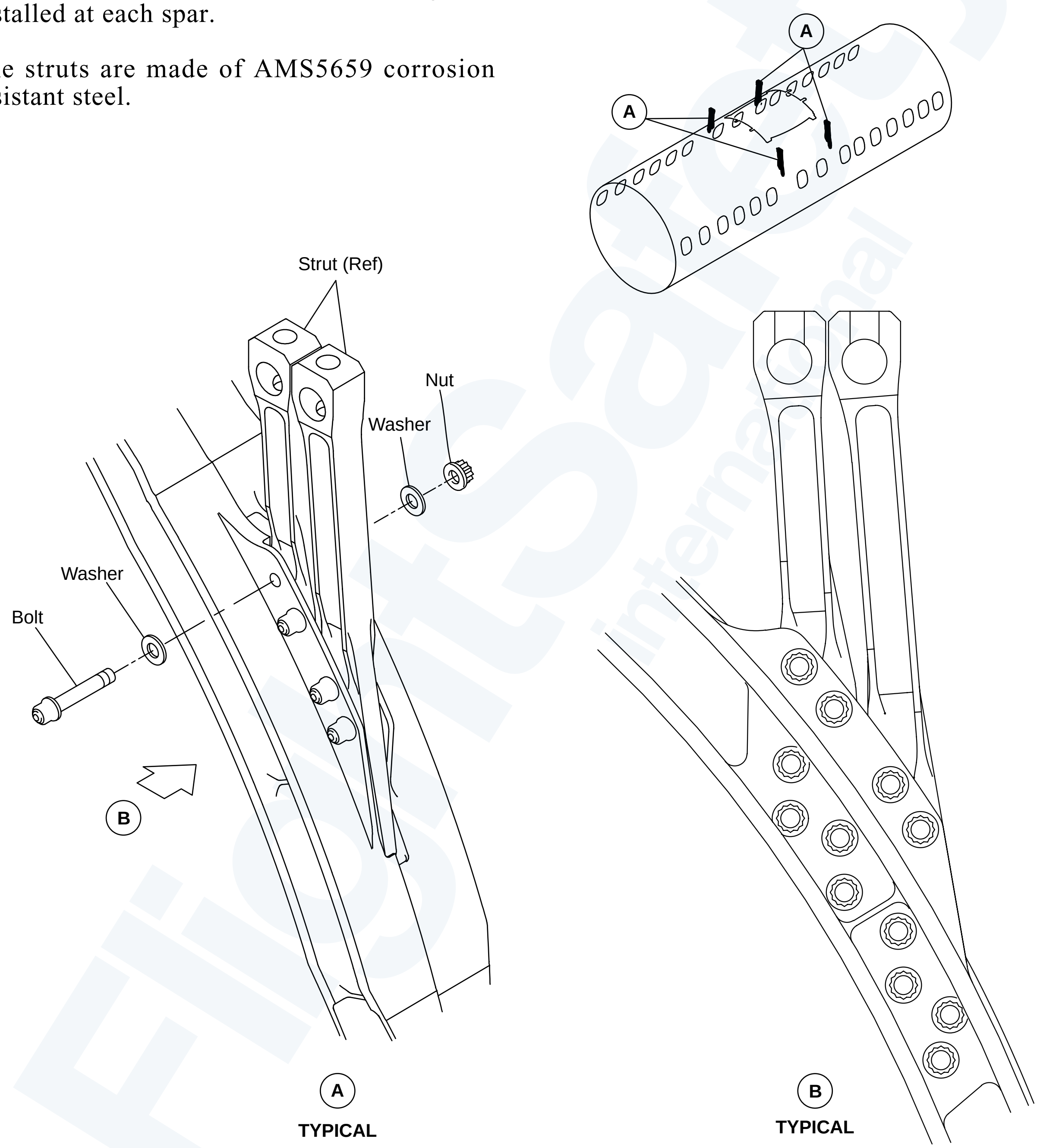

Figure 53-18. Wing to Fuselage Joint Struts

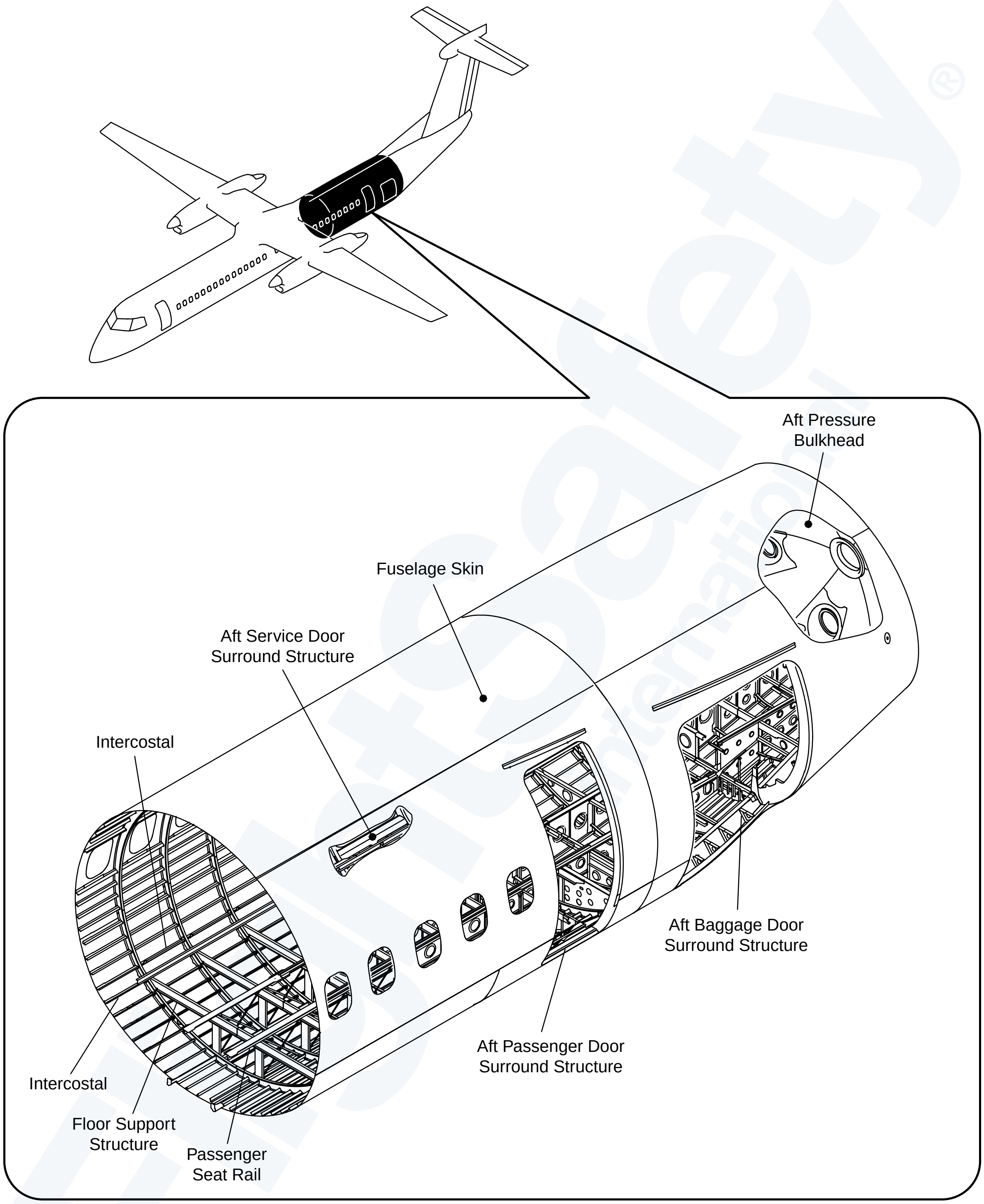

Figure 53-19. Aft Center Fuselage

53-40-00 AFT CENTER FUSELAGE

INTRODUCTION

The aft center section extends rearward to the fuselage aft center section, from station (X566.025 to X829.548/836.452) along the longitudinal X-axis.

GENERAL DESCRIPTION

Refer to Figure 53-19. Aft Center Fuselage.

The aft center structure has the components that follow:

- Stringers
- Frames
- Intercostals
- Aft pressure bulkhead
- Aft service door surround structure
- Aft passenger door surround structure
- Fuselage skin
- Strakes
- Floor support structure
- Floor panels
- Passenger seat rail
- Aft baggage door surround structure
- Baggage compartment liner.

NOTE

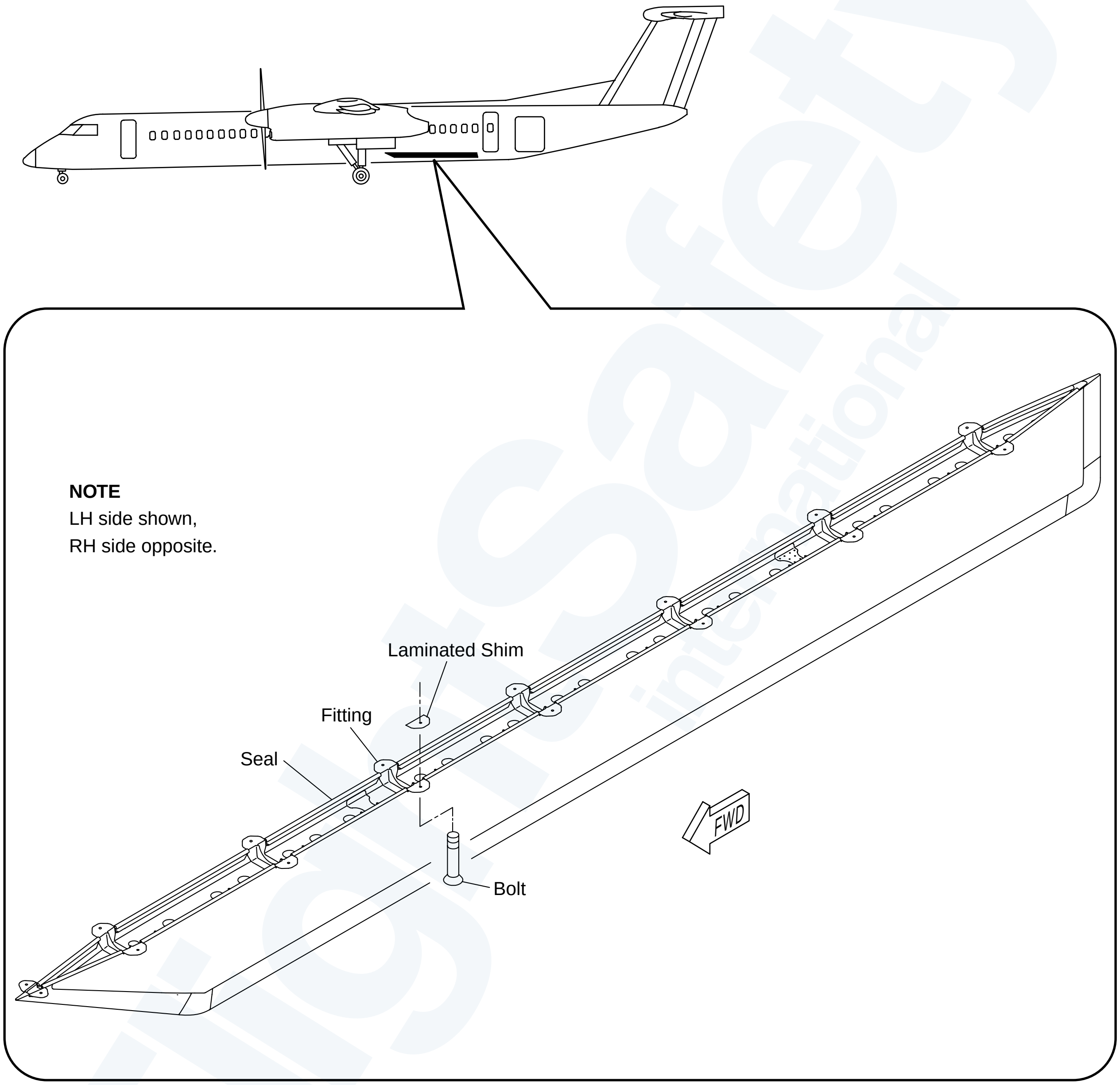

Figure 53-20. Fuselage Strakes

COMPONENT DESCRIPTION

Refer to Figure 53-20. Fuselage Strakes.

The strakes are installed below the center fuselage.

The strake structure has the following components:

- Skin
- Fairing caps
- Fittings
- Fitting Angles
- Channels
- Splices
- Caps
- Spars
- Ribs.

The skin is made of Alclad 2024-T3.

The fuselage strakes are aerodynamic components of the aircraft. They give longitudinal stability when the aircraft is in flight.

NOTE

Forward Dorsal Fairing

Refer to Figure 53-21. Forward Dorsal Fairing.

The aft center fuselage section has the forward dorsal panels that follow:

- Dorsal fin fairing 351BT
- Dorsal fin fairing 351CT
- Dorsal fin fairing 351DT
- Dorsal fairing panels 352AL/352AR
- Dorsal caps zone 352.

The forward dorsal fin structural components and skin has the following materials:

- Epoxy pre-impregnated glass cloth
- Honeycomb core
- Moisture barrier
- Pre-impregnated cloth-aramid (aromatic polyamide) fabric epoxy.

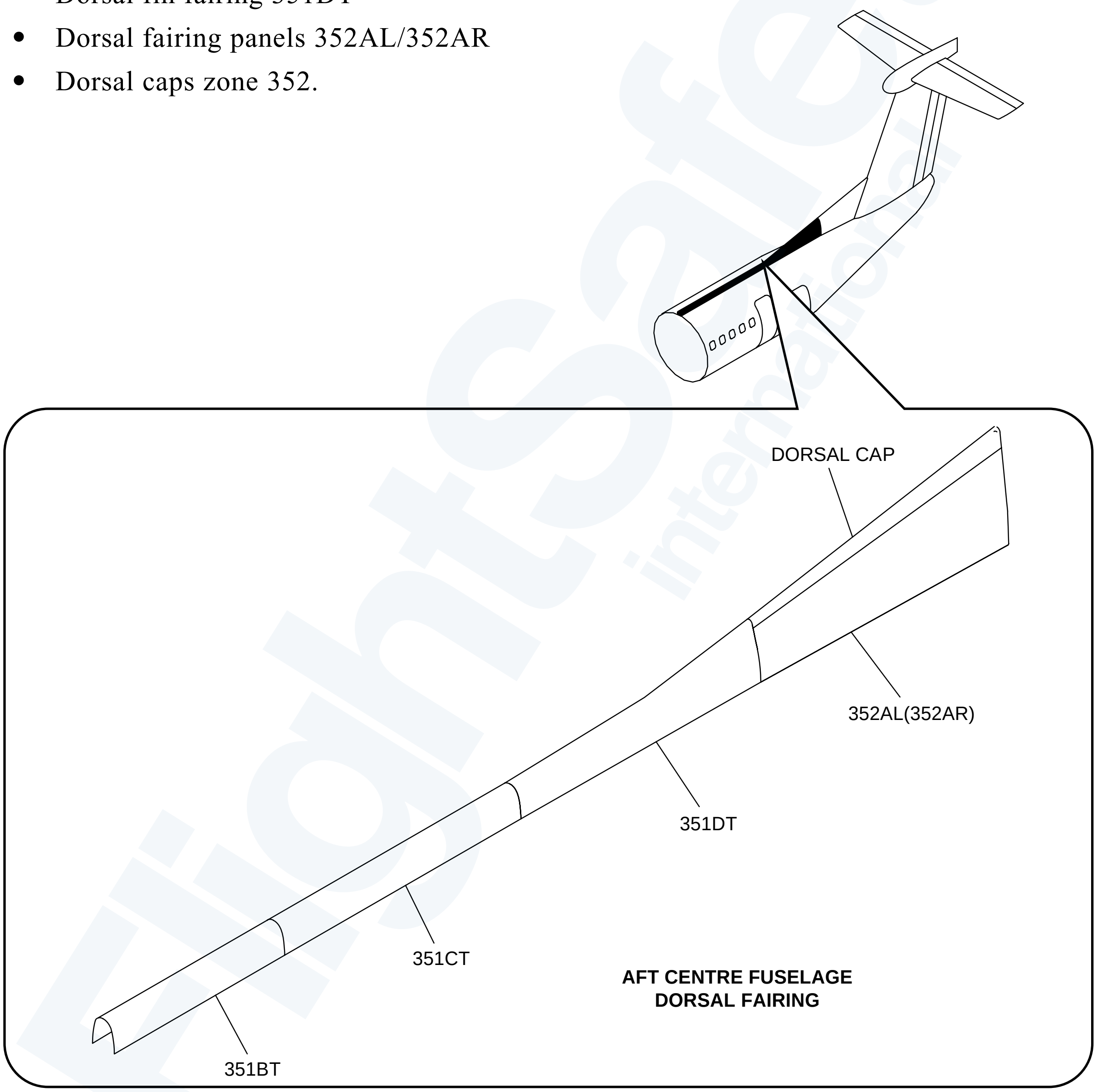

Figure 53-21. Forward Dorsal Fairing

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Aft Dorsal Fairing

Refer to Figure 53-22. Aft Dorsal Fairing.

The aft center fuselage section has the aft dorsal panels that follows:

Aft Fuselage Dorsal fairing, zone 321.

The aft dorsal fin structural components and skin has the materials that follow:

- Fibre mat
- Epoxy pre-impregnated glass cloth
- Pre-impregnated cloth-aramid fabric epoxy
- Moisture barrier
- Honeycomb core.

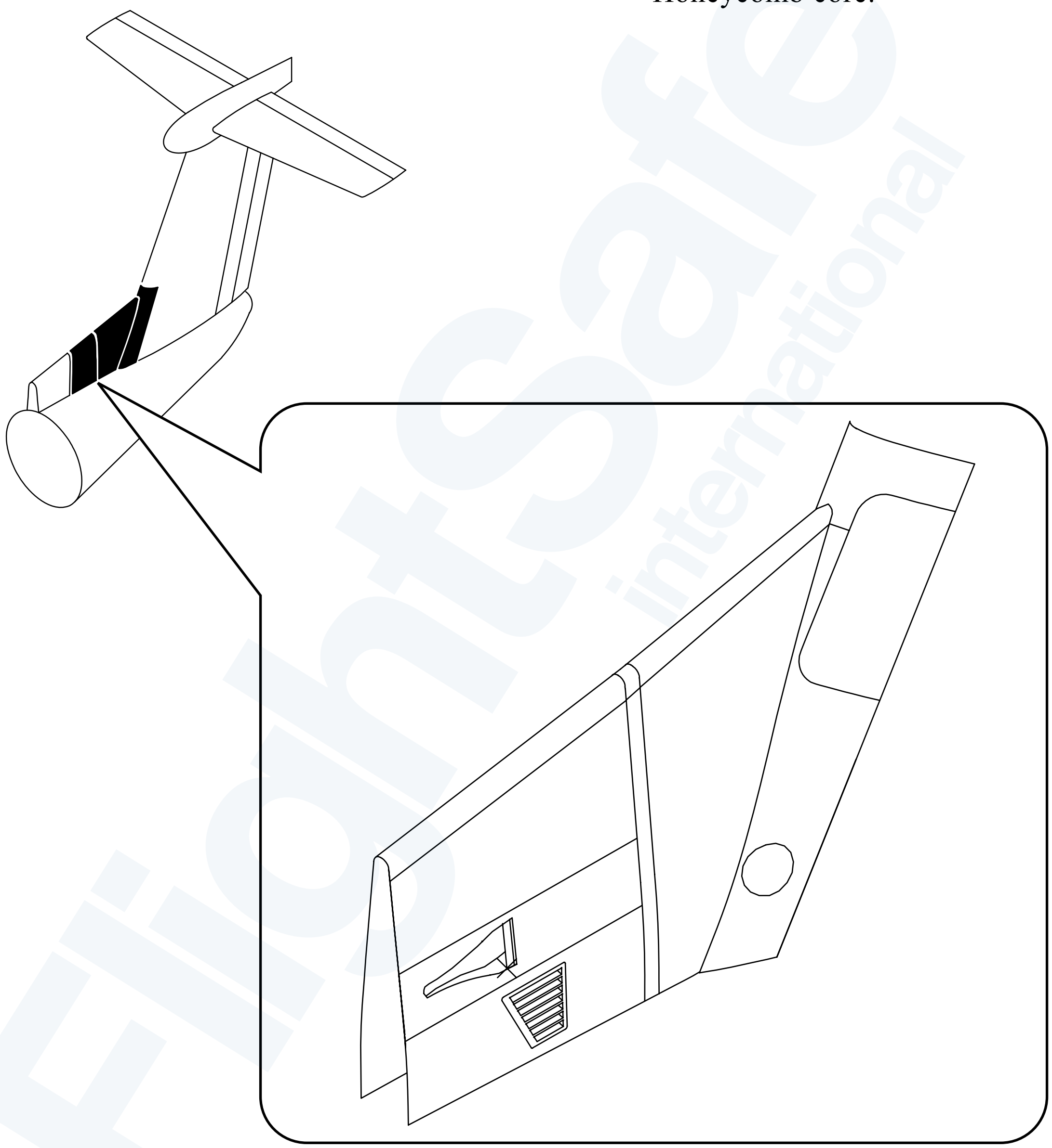

Figure 53-22. Aft Dorsal Fairing

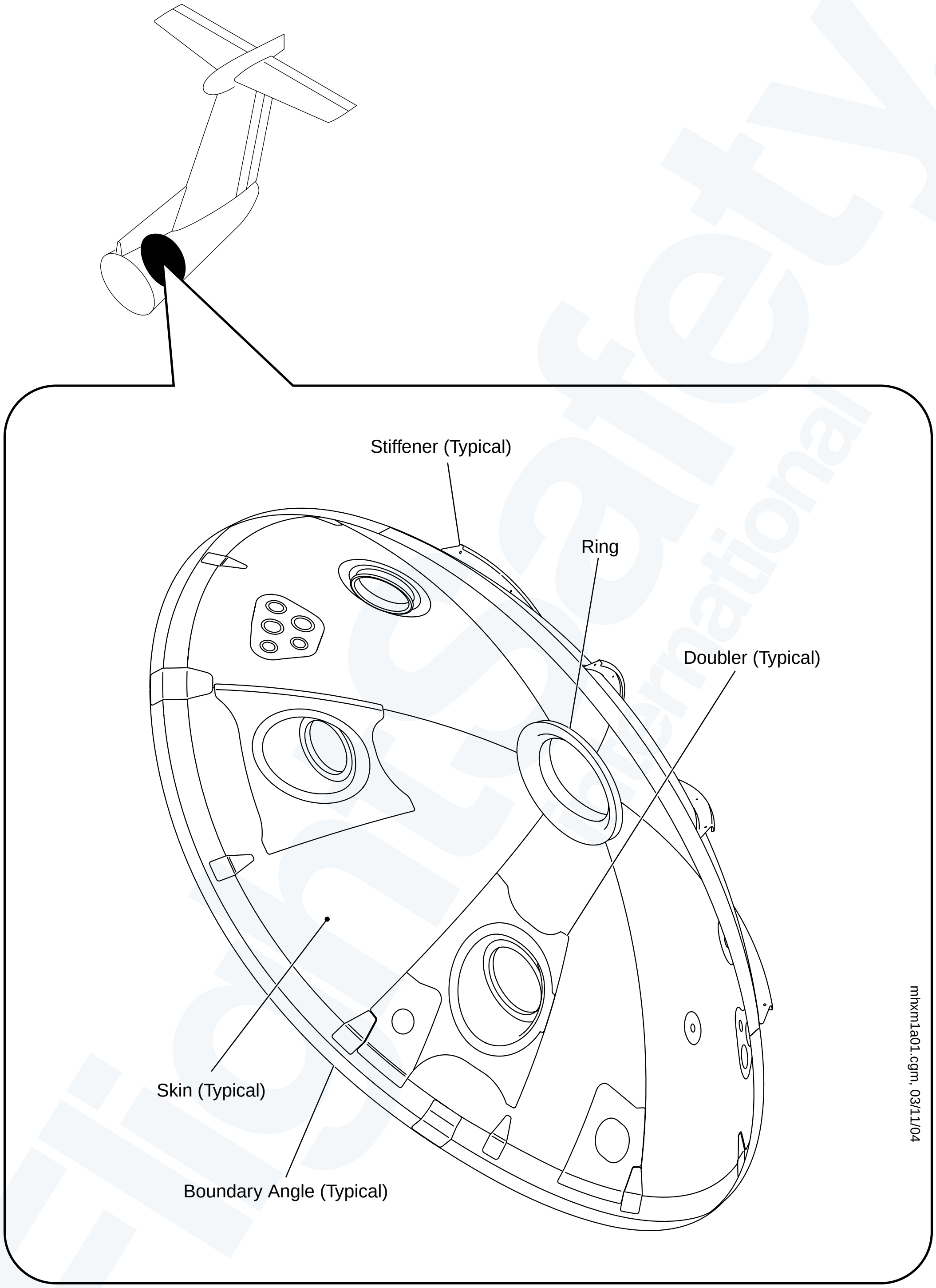

Figure 53-23. Detailed Visual Inspection of the Rear Pressure Dome, Pressure Dome Structure (ALI#532027F101)

53-40-61 REAR PRESSURE DOME STRUCTURE

Refer to Figure 53-23. Detailed Visual Inspection of the Rear Pressure Dome, Pressure Dome Structure (ALI#532027F101).

Basic pressure dome structure: There are skins, stiffeners, boundary angles, ring and doublers in the rear pressure dome.

DVI: Examine the skins, stiffeners, boundary angles, ring and doublers. Make sure that there is no damage, cracks, or corrosion.

NOTE

Use a light source, inspection mirror and magnifying glass as necessary.

No repairs are allowed to the Pressure bulkhead.

If damage, cracks, or corrosion is found on the forward and aft surfaces of the rear pressure dome, contact Bombardier Aerospace In-Service Engineering Structures for a disposition.

NOTE

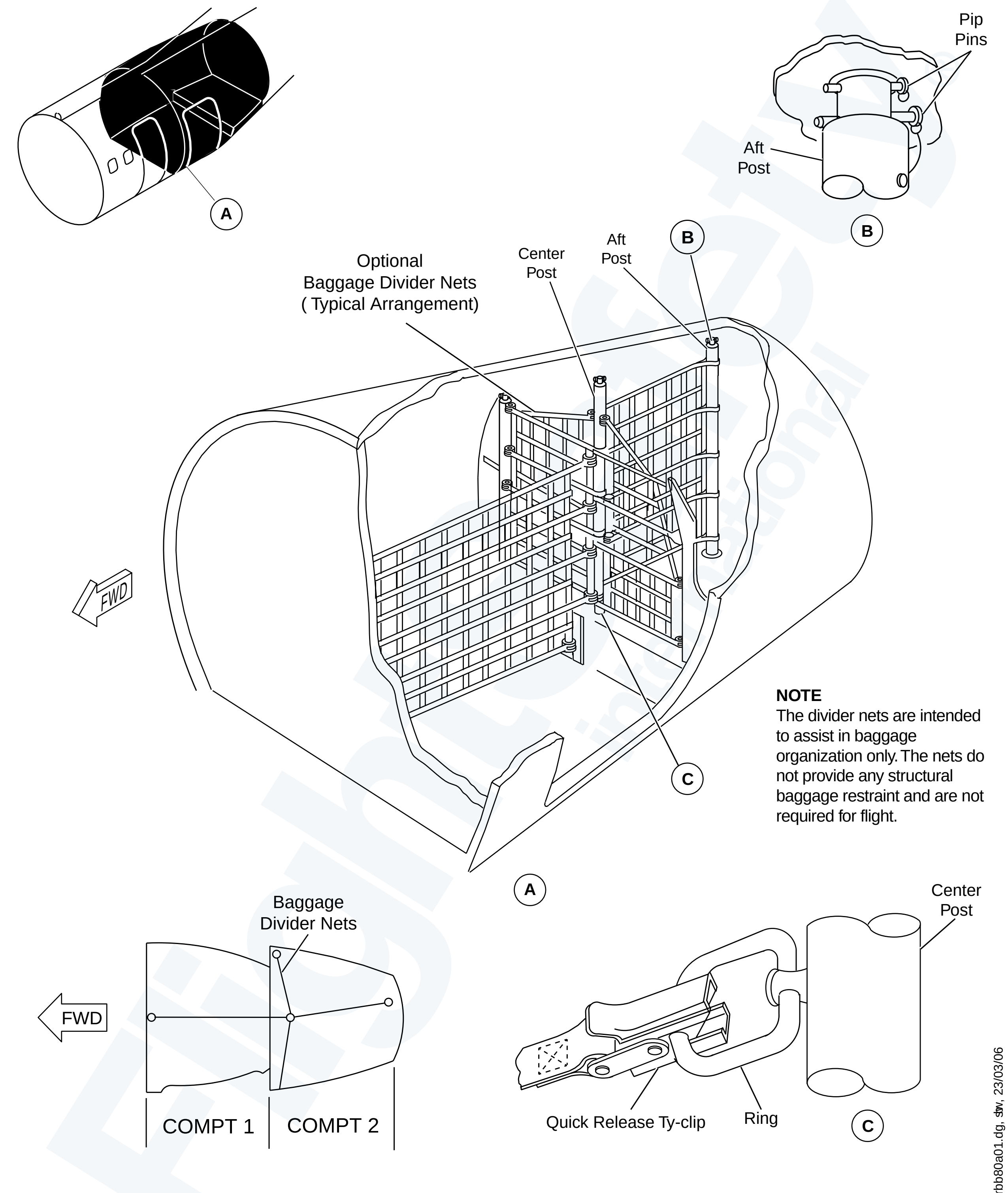

Figure 53-24. Baggage Divider Nets - Aft Baggage Compartment

Baggage Compartment Divider Nets

Refer to Figure 53-24. Baggage Divider Nets - Aft Baggage Compartment.

Baggage restraints, optional baggage divider nets and tie-down rings are installed in the aft baggage compartment to restrain cargo and baggage, and keep the cargo and baggage from moving during flight.

CAUTION

THE BAGGAGE DIVIDER NETS ARE INTENDED TO BE USED TO HELP ORGANIZE BAGGAGE ONLY. DO NOT USE BAGGAGE DIVIDER NETS TO PROVIDE STRUCTURAL RESTRAINTS. IF YOU USE THE BAGGAGE DIVIDER NETS AS STRUCTURAL RESTRAINTS, THEN YOU MAY CAUSE DAMAGE TO EQUIPMENT.

Temporary Allowance for Continued Operation with One Missing or Damaged Attachment Clip on the Aft Baggage Compartment Restraint Net 25 Hours Max

The repair consists of inspection of the restraint net, removal of the damaged attachment clip from the cargo net and securing the loose strap.

Baggage Compartment Liner

Refer to Figure 53-25. Baggage Compartment.

The cargo liner is installed with screws, in multiple pieces and sealed with fire resistant tape and RTV.

The liner is sealed to provide a protective layer designed to give some protection to the structure around the cargo area from fire, abrasion and impact.

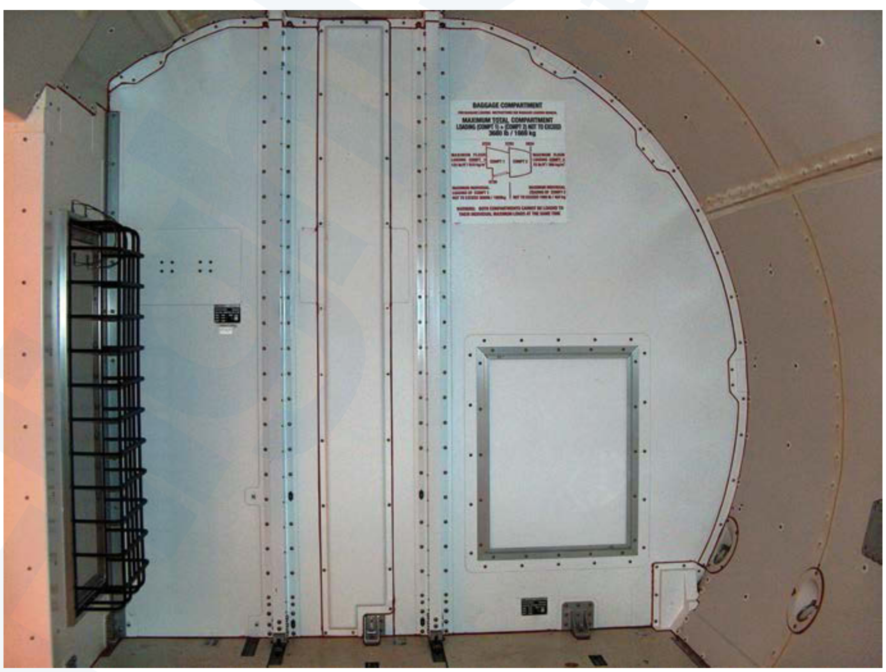

Figure 53-25. Baggage Compartment

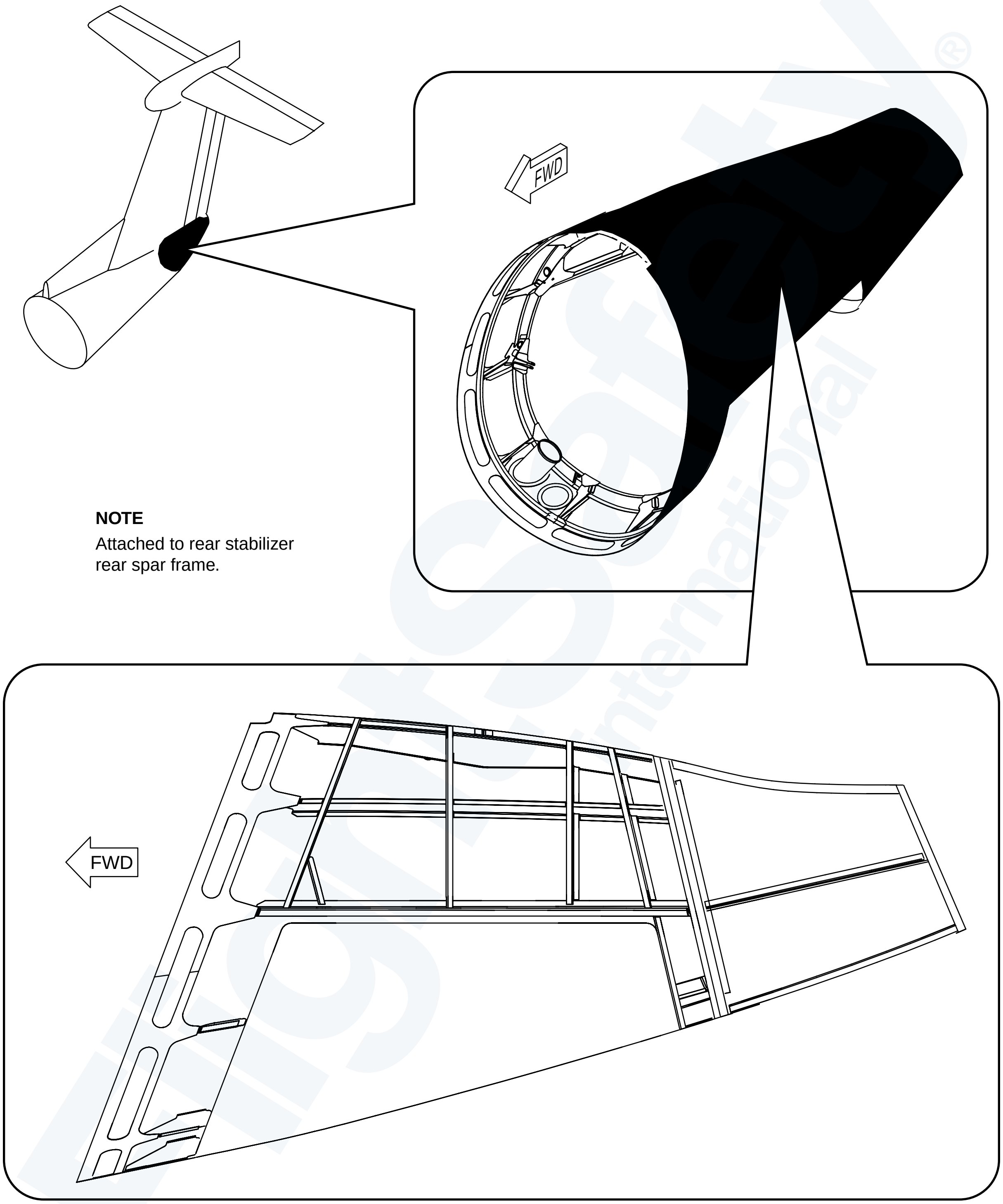

Figure 53-26. Aft Fuselage Tailcone

53-60-00 AFT FUSELAGE TAILCONE

INTRODUCTION

The tailcone is attached to the rear fuselage at the location of the vertical stabilizer rear-spar frame. The tailcone structure is designed to support the vertical load reaction from the aft auxiliary power unit mount, as well as inertia loading from the weight of the structure and external aerodynamic pressure loading. The structure is also designed to support the vertical load due to lifting and hoisting of the auxiliary power unit.

GENERAL

Refer to Figure 53-26. Aft Fuselage Tailcone.

The tailcone structure is divided into a forward and aft assembly at canted station 1033.46. The structure of the forward tailcone has a thin skin external shell that is reinforced by longitudinal intercostals and transverse frames. The skin is titanium.

The tailcone assembly has a firewall 5 inches aft of the rear spar frame of the vertical stabilizer. The fire containment zone of the auxiliary power unit is enclosed by the tailcone firewall.

Aft fuselage tailcone has the following components that follow:

- Skin
- Frame
- Intercostals
- Firewall
- Seal and strap
- Hinge lug fittings.

NOTE

53-00-00 APPENDIX

NOTES

MAINTENANCE CONSIDERATION

Unscheduled Inspection

Refer to the Bombardier published AMM Part 2 PSM 1-84-2.

- TASK 05-50-08-210-801 Inspection after a Tail Strike
- TASK 05-50-11-210-801 Inspection after Extreme Turbulence or Buffeting
- TASK 05-50-16-210-801 Inspection after a Bird Strike Condition
- TASK 05-50-21-210-801 Inspection after a Lightning Strike
- TASK 05-50-21-210-802 Inspection after a Lightning Strike for a Ferry Flight Only
- TASK 05-50-26-210-801 Inspection after Operation at More than the Maneuvering Limit Load Factors
- TASK 05-50-28-210-801 Inspection after Operation at More than the Maximum Operating Speed
- TASK 05-50-31-210-801 Inspection after Operation at More than the Maximum Gear Operation Speed
- TASK 05-50-36-210-801 Inspection after Operation at More than the Wing Flaps Down Speed
- TASK 05-50-41-210-801 Inspection after the Aircraft is Parked In Strong Winds or Wind gusts
- TASK 05-50-46-210-801 Inspection after the Application of Too Much Ground Deicing Fluid
- TASK 05-50-06-210-801 Inspection after a Hard Landing
- TASK 05-50-51-210-801 Inspection after Rough Towing.

53-00-00 SPECIAL TOOL & TEST EQUIPMENT

- Commercially Available Calibration Test Standard (CTS), P/N SRS-0824A (NDT Eng. Corp.)
- GSB2000001 Borescope - 110 Volt, 60 Hz (if necessary)
- GSB2000015 Borescope - 220 Volt, 50 Hz (if necessary)
- Commercially Available (or manufacture it in-house per NDTM task) Calibration Test Standard for HFEC Inspection, Nonferrous (NFe) coupon made of aluminum alloy 7075/T6, bare, with electrical discharge machining (EDM) notches
- Commercially Available Calibration Test Standard for HFEC Inspection, P/N 29A047 (Hocking)
- Commercially Available Calibration Test Standard (CTS) for LFEC inspection, P/N EXBD400-50071-001 (Bombardier Aerospace, Toronto)
- Commercially Available Calibration Test Standard (CTS) for HFEC inspection, P/N SRS 0824A (NDT Engineering Corporation)
- GSB0700025 (or equivalent) Engine straps
- Commercially Available 25X Torque Multiplier
- Commercially Available Load cell (can measure a 2500 lbs (1134 kg) load)
- Commercially Available Mechanical or hydraulic lifting device (can operate manually)
- GSB0700024 Stand - Tail Support
- Available from Bombardier Aerospace (Toronto) Calibration Test Standard (CTS), P/N EXBD400-53070-001
- Available for rent from Bombardier Aerospace (Toronto) Calibration Test Standards for HFEC and LFEC inspections, P/N 99A14144-101 and 99A14145-101
- Available from Bombardier Aerospace (Toronto) Calibration Test Standards (CTS), P/N 99A14144-101, P/N 99A14145-101 and P/N EXBD400-54006-001

53-00-00 MAINTENANCE PRACTICES

Refer to the Bombardier AMM PSM 1-84-2 for details on these maintenance procedures:

- AMM 53-40-51-000-802: Removal of the Main−Hinge Attach Fittings - Aft Entry Door.
- AMM 53-40-51-400-802: Installation of the Main−Hinge Attach Fittings - Aft Entry Door.